DETAILED PROJECT REPORT

AI Video Generation Platform

India-Centric, AI Video Generation and Production Pipeline

September 2026

Consolidated from eight module-level research documents

Field	Entry
Project	AI Video Generation Platform
Document type	Detailed Project Report (DPR) – implementation report
Pipeline	Prompt → Plan → Image → Video → Render → QA → Store → Deliver
Project duration	3 months (planned October – December 2026)

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1. Executive Summary

The proposed platform is an end-to-end AI video generation and production system that converts a natural-language user request into a structured production plan, generates visual and , validates consistency and quality, synchronizes all media on a common timeline, renders the final output and stores it for controlled delivery.

The central architectural principle is modular orchestration rather than dependence on one model. Prompt Intelligence converts the request into a validated VideoSpec. Planning converts that specification into a VideoBlueprint and dependency graph. Image generation establishes references and keyframes; video generation produces motion; Module 5 acts as a quality gate; the rendering layer composes approved assets; and the backend/infrastructure layers provide persistence, asynchronous execution, security and scalable compute.

The research places particular emphasis on consistency. Long or multi-scene generated videos can suffer from identity drift, scene changes, object changes, temporal instability, motion anomalies. The QA design therefore combines identity embeddings, general visual embeddings, perceptual and structural similarity, optical flow, semantic alignment and artifact inspection, with PASS, AUTO_RETRY and HUMAN_REVIEW outcomes.

The infrastructure direction is hybrid: lightweight tasks can run locally or on CPU/managed services, while large video generation can use high-VRAM cloud GPUs or managed APIs. FastAPI, PostgreSQL, Redis/Celery, object storage, GPU workers and FFmpeg provide the proposed application backbone.

Integrated Value Chain

- Natural language becomes a machine-readable production specification.
- Planning separates story, scenes, shots, timing and dependencies before expensive generation.
- Reference-conditioned image generation establishes stable visual anchors.
- Video generation converts keyframes or prompts into motion clips.
- QA evaluates visual, temporal, identity and semantic consistency before final rendering.
- Rendering uses a canonical timeline and media inspection to produce a validated final file.
- Backend and infrastructure provide job tracking, ownership, storage, monitoring and scaling.

2. Introduction and Background

2.1 Project Context

AI video creation is not a single-model problem. A complete production workflow combines language understanding, planning, image generation, motion generation, temporal composition, quality control and infrastructure. The source research consistently separates these concerns so each can be tested and replaced without redesigning the entire platform.

2.2 Core Production Abstraction

The consolidated design uses a hierarchy of prompt → VideoSpec → VideoBlueprint → scenes/shots → assets → timeline events → final export. VideoSpec captures what the user wants. VideoBlueprint describes how the system intends to produce it. Assets are the generated media objects. Timeline events map assets to time. The final render is the assembled representation.

Abstraction	Purpose	Produced by	Consumed by
Prompt	Original user intent	User / frontend	Prompt Intelligence
VideoSpec	Validated structured requirements	Prompt Intelligence	Planning + downstream modules
VideoBlueprint	Scene/shot/timing execution plan	Planning	Generation + rendering
Timeline	Ordered time-based composition	Planning + rendering	Gstreamer/FFmpeg + export
Manifest / QA report	Provenance and validation evidence	QA / rendering	Backend / user / operations

2.3 India-Centric Requirement

The project research includes Indian faces and project-defined regional/persona appearance groups. The design therefore treats identity references, regional/persona validation as first-class metadata rather than optional presentation details.

3. Problem Statement, Objectives and Scope

3.1 Problem Statement

Creating a coherent educational, storytelling or cinematic video normally requires topic interpretation, script preparation, scene planning, visual design, character consistency, motion generation, synchronization, rendering and publishing. A single text-to-video request does not reliably solve the full production problem, especially when a video contains multiple scenes, recurring characters or independent and visual events.

3.2 Objectives

Accept natural-language prompts and convert them into validated structured requirements.
Create a deterministic, machine-readable production blueprint with scene and shot dependencies.
Generate reference images and keyframes that establish character, object and environment appearance.
Generate motion clips through text-to-video and image-to-video/keyframe-conditioned approaches.
Detects identity drift, visual artifacts, motion anomalies, flicker and semantic mismatches.
Synchronize independent media streams on a canonical timeline.
Provide secure job tracking, persistent metadata, private media storage and controlled delivery.
Support hybrid deployment and model routing according to workload, VRAM, latency, cost and licensing.

3.3 Scope

- In scope: prompt intelligence, planning, image generation, video/motion generation, visual QA, timeline/rendering, backend/API services and infrastructure.
- Operational capabilities: asynchronous jobs, retries, cancellation, ownership checks, storage lifecycle, logging and monitoring.
- Out of scope or unresolved: final provider selection, exact production thresholds, final commercial licenses, final concurrent-job target and definitive production cost.

4. Existing System

Google Veo – Existing AI Video Generation System

Google Veo is Google DeepMind's generative video model family for creating video from natural-language prompts and visual inputs. The current Veo 3.1 generation supports text-to-video and image-to-video workflows, native reference-image guidance, video extension, first-and-last-frame generation, and other creative controls. Veo can be accessed through Google products such as Gemini and Flow and can also be integrated programmatically through the Gemini API.

4.1 Key Capabilities of Google Veo

Text-to-Video Generation: Generates video from natural-language prompts describing the scene, action, camera movement, style and other visual requirements.

Image-to-Video Generation: Uses an input image as the starting visual frame and generates motion and video content around it.

Reference Images: Veo 3.1 supports up to three reference images to guide the appearance of a person, character or product and improve visual consistency.

First and Last Frame Control: Allows a starting image and ending image to be specified so that the generated video transitions between defined visual states.

Video Extension: Previously generated Veo video can be extended to continue a scene while attempting to maintain visual continuity.

Camera and Motion Controls: Provides controls for framing, camera movement and object motion to provide greater control over generated shots.

Multiple Resolutions and Aspect Ratios: Veo 3.1 supports 720p, 1080p and 4K generation under model-specific constraints, along with landscape 16:9 and portrait 9:16 formats.

API Integration: Veo 3.1 can be accessed through the Gemini API, allowing video generation to be incorporated into an external software pipeline.

4.2 Existing System Workflow

A typical Veo-based generation workflow can be represented as:

Prompt / Image Input → Veo Generation → Video → Post-Processing / Delivery

For API-based use, the application can submit a prompt and, where applicable, an input image or reference images to the Veo model. The generated video is returned after the asynchronous generation operation is completed. An external application can then perform additional operations such as storage, quality validation, editing, timeline composition, branding and final delivery.

4.3 **Technical Specifications Relevant to the Proposed System**

Parameter	Google Veo 3.1
Input	Text and image; reference-image support is available in Veo 3.1
Generation Type	Text-to-video and image-to-video; video extension is also supported
Reference Images	Up to 3 reference images for content/style guidance
Video Duration	4, 6 or 8 seconds, with specific feature/resolution constraints
Resolution	720p, 1080p and 4K for applicable Veo 3.1 generation modes
Aspect Ratio	16:9 and 9:16
Frame Rate	24 fps
API	Gemini API

4.4 Strengths of the Existing System

- Strong text and image conditioning for cinematic video generation.
- Reference-image capability can help maintain the appearance of characters, people and products across generated shots.
- First/last-frame and extension capabilities provide additional control for sequential visual content.
- Programmatic API access makes the model suitable as a generation component inside a larger application.

4.5 Limitations / Gaps for the Proposed Platform

- Veo is primarily a video-generation model rather than a complete end-to-end production management platform.
- A multi-scene advertising workflow still requires external orchestration for prompt decomposition, scene planning, asset management, QA, timeline management, rendering and delivery.
- Veo generates short clips per generation request; therefore, a longer multi-scene advertisement requires application-level sequencing and synchronization.
- Application-level cost control, retry policies, model routing, audit logging and production governance must be handled by the surrounding platform.

4.6 Relevance to the Proposed AI Video Generation Platform

Google Veo can be considered an existing foundation-model capability that the proposed platform can use as a video-generation engine. The proposed system is intended to address the higher-level production workflow around such a model. In particular, the proposed architecture can add prompt intelligence, multi-scene video planning, image/keyframe generation, visual consistency controls, automated QA, timeline synchronization, rendering, storage and delivery. This creates a separation between the underlying generative model and the application-level production orchestration layer.

Proposed relationship: User Input → Prompt Intelligence → Video Planning → Image/Keyframe Generation → Veo/Other Video Models → QA → Timeline/Render → Delivery

The above description is based on Google DeepMind's current Veo documentation and Google's Gemini API documentation for Veo 3.1, accessed in September 2026. Model capabilities and availability may change as Google releases newer Veo versions.

5. Proposed System

The proposed platform is an asynchronous, modular media-generation system. The frontend communicates with a versioned FastAPI contract. The backend persists project and job state before dispatching work. Celery workers consume jobs through Redis, generation adapters call local models or external providers, QA evaluates outputs, and the rendering worker produces the final artifact.

5.1 Proposed System Characteristics

- **Modular Architecture**: Separate modules handle prompt processing, video planning, image generation, video generation, QA, rendering, backend, and deployment.
- **Structured Workflow**: Converts the user's prompt into a structured video specification and production plan.
- **Asynchronous Processing**: Long-running generation and rendering tasks are handled as background jobs.
- **Dependency-Aware Execution**: Each stage starts only after its required inputs from the previous stage are available.
- **Reference-Conditioned Generation**: Character, object, environment, and style references are maintained across scenes.
- **Quality-Gated Generation**: Generated clips are checked for visual quality, consistency, artifacts, and temporal stability before acceptance.
- **Automatic Retry**: Failed quality checks can trigger controlled regeneration within a defined retry limit.
- **Provider-Agnostic**: Model adapters allow different AI image and video generation models to be integrated or replaced.
- **Separated Storage**: Project metadata is stored in the database, while images, video clips, and final outputs are stored separately.
- **Cost-Aware Processing**: Model selection, previews, controlled retries, and resource scheduling help reduce unnecessary processing costs.
- **Scalable Processing:** Queues and worker services allow multiple video-generation jobs to be processed efficiently.
- **Traceable Workflow**: Prompts, references, generated assets, QA results, job status, and final outputs are recorded for monitoring and reproducibility.

5.2 Proposed Lifecycle

User Login → User Dashboard → Enter Prompt → Avatar Selection / Character Generation → Character ID Creation → JEV Processing → Gemini 3.8 Flash Scene Orchestration → Video Blueprint Creation → Wan 2.2 Video Generation → Character ID Tracking & Consistency Check → FFmpeg Final Rendering → Final Quality Check → Save Video → Deliver to User

6. System Requirements

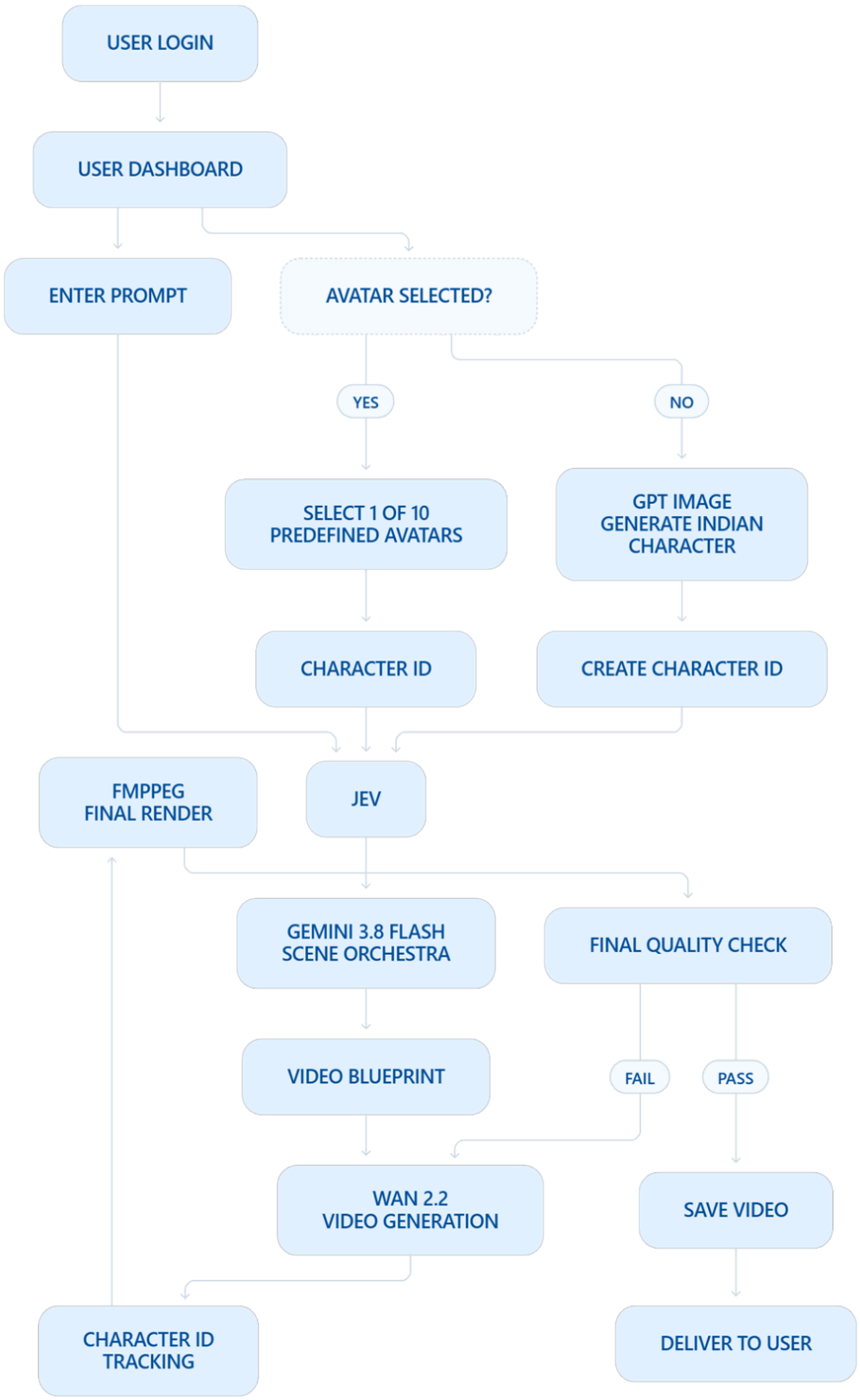
6.1 Functional Requirements

ID	Requirement	Primary Owner
FR-01	Accept natural-language prompt and settings	Frontend / Prompt
FR-02	Detect language, intent, entities, actions and visual requirements	Prompt
FR-03	Generate and validate VideoSpec JSON	Prompt
FR-04	Create VideoBlueprint, scenes, shots and dependencies	Planning
FR-05	Generate references/keyframes with conditioning	Image
FR-06	Generate video/motion clips	Video
FR-07	Perform identity, visual, temporal and semantic QA	QA
FR-08	Regenerate or route low-confidence outputs	Planning + QA
FR-09	Compose assets on a canonical timeline	Rendering
FR-10	Store job state, assets, manifests and audit metadata	Backend
FR-11	Authenticate users and enforce project ownership	Backend
FR-12	Queue and schedule long-running GPU/API tasks	Infrastructure
FR-13	Export a validated downloadable video	Rendering
FR-14	Track progress, retries, cancellation and failures	Backend / Orchestration

6.2 Non-Functional Requirements

- Reliability: idempotent worker writes and provider callbacks; explicit failure states.
- Security: authentication, authorization, private media, safe file handling and controlled credentials.
- Scalability: API replicas and GPU workers scale independently.
- Observability: p50/p95 latency, queue wait, GPU utilization, failure/retry rate and cost per accepted output.
- Reproducibility: store prompt version, seed, model/version, settings and render manifest where applicable.
- Maintainability: model adapters, versioned schemas and isolated module responsibilities.

7. Consolidated System Architecture



8. Module Architecture and Responsibilities

The architecture separates the control plane (API, authentication, contracts, orchestration) from the media/data plane (generation workers, QA, rendering, database and object storage). This separation allows expensive media processing to proceed asynchronously without blocking user-facing requests.

8.1 Architectural Principles

1. **User Prompt** – User provides the topic, story, characters, and video requirements.
2. **Story Planner** – Converts the user prompt into a structured story, scenes, and shots.
3. **Character / Avatar** – Creates or selects the required characters and maintains their appearance.
4. **Scene Generation** – Creates scene descriptions, keyframes, and reference images.
5. **WAN Animate-14B** – Generates video using:
    - **T2V A14B:** Text → Video
    - **I2V A14B:** Image → Video
6. **WAN VACE** – Provides video editing and control for the generated clips.
7. **Editing / Control** – Arranges shots, timing, transitions, and visual adjustments.
8. **GStreamer Rendering** – Combines and renders the video into the required output format.
9. **FFmpeg Fallback** – Used when GStreamer cannot perform a required rendering operation.
10. **Rendered Video** – Produces the processed video that is ready for quality checking.
11. **QA Engine** – Checks the generated video using multiple quality models:

- **AdaFace:** Character identity consistency
- **DINOv3:** Visual/object consistency
- **RAFT:** Motion and temporal stability
- **CLIP:** Text-to-video/semantic alignment
- **LPIPS/SSIM:** Visual quality and similarity

12. **VBench** – Performs overall video-quality evaluation.
13. **Quality Gate** – Decides whether the video meets the required quality standards.
14. **PASS** – Approved video becomes the **Final Output**.
15. **FAIL** – Video is sent for **Regeneration or VLM Escalation/Review**.

9. Module Architecture and Responsibilities

Module	Inputs	Core Processing	Outputs
1. Prompt/NLP	User prompt	Language, intent, entities, actions, visual settings extraction; enhancement; safety; validation	VideoSpec
2. Planning/ Orchestration	VideoSpec	Scenes, shots, durations, DAG, jobs, retries	VideoBlueprint + job graph
3. AI Image	Blueprint + references	T2I/I2I,V2V identity/style conditioning, keyframes, metadata	Reference images + keyframes
4. AI Video/Motion	Prompts + keyframes	T2V/I2V,V2V motion/camera control, clip generation	Video clips + generation metadata
5. Visual QA, Accountability	References + clips + metadata	Identity, visual, temporal, semantic, artifact checks,,ownership,traceability,content protection	QA report + decision
6. Timeline/Rendering	Blueprint + approved assets	Preflight, normalization, timeline, mix, subtitles, encode	Final video + manifest
7. Backend	Users/projects/jobs/assets	API, auth, DB, storage, job state	Persistent system state
8. Infrastructure	All compute workloads	GPU/CPU resources, queues, containers, scaling, monitoring	Execution environment

Every module should document a stable input schema, output schema, correlation/job identifier, version, ownership rule and error behavior. This follows the backend research emphasis on explicit contracts and makes the nine modules function as one system.

9.1 Module 1 — Prompt Intelligence & NLP

9.1.1. Production Role and Design Decision

Production decision: use Gemini 3.8 Flash as the generative/reasoning workhorse and Jev as the fast typed decision and gating layer. The supplied Module 1/2 research explicitly positions Gemini for extraction, structured generation and reasoning while Jev handles high-volume safety, ambiguity, routing, scoring and pass/fail decisions.

9.1.2. Why Gemini 3.8 Flash + JevGemini 3.8

Flash is assigned the higher-complexity language work: prompt interpretation, structured extraction, scene intent and VideoSpec drafting. Jev is assigned the fast, typed decision path for safety, ambiguity, completeness, routing and confidence checks. This separation reduces unnecessary expensive model calls and keeps the control path deterministic. A practical production sequence is: Jev pre-check → Gemini structured extraction → Jev validation/gating → Pydantic schema validation → persisted VideoSpec. The 0.85 confidence value remains a calibration starting point from the supplied research, not a permanent universal threshold.

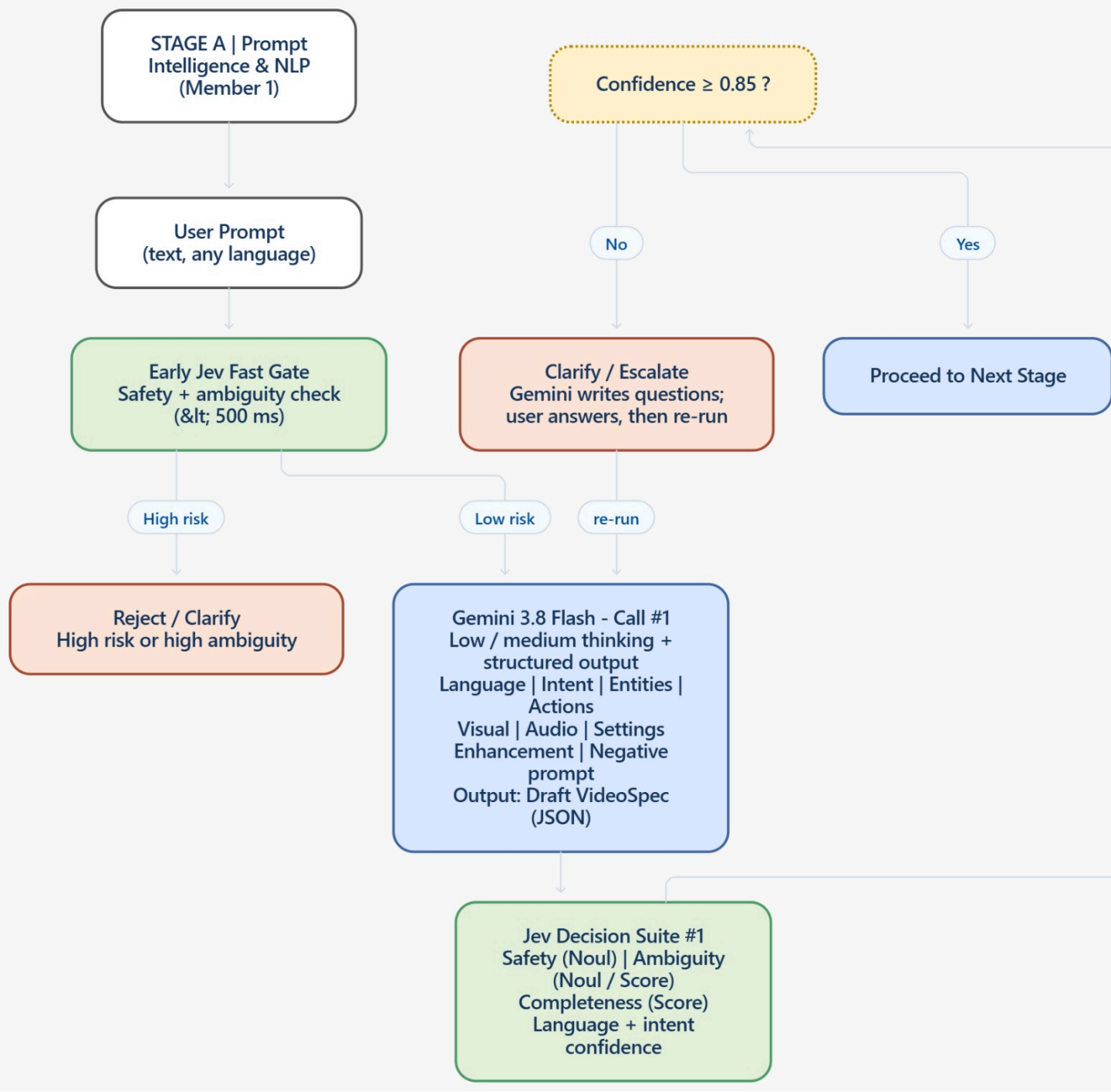
9.1.3. Production Model / Technology Comparison

Criterion	Selected	Why selected	Alternative / trade-off
Structured understanding	Gemini 3.8 Flash	Handles language, intent, entities, actions, visual settings extraction and structured JSON.	Lighter models reduce cost but may need more application logic.
Reasoning/enhancement	Gemini 3.8 Flash	Supports multi-step prompt enhancement without changing user intent.	Jev is not used for free-form generation.
Safety/ambiguity	Hybrid	Typed Noul/Choice/Score outputs and confidence gates reduce free-text decision risk.	Gemini-only uses more tokens for repeated small decisions.
Routing/scoring	Jev	Designed for parallel high-volume decisions.	General LLM routing is more expensive.
Escalation	Hybrid	Low Jev confidence can trigger Gemini clarification or human review.	Single-model designs separate generation and gating less clearly.

9.1.4. Production Implementation Controls

- Receive prompt through a versioned API with user/project/correlation IDs.
- Run early Jev safety and ambiguity gate before expensive generation.
- Call Gemini with structured JSON schema for extraction and VideoSpec drafting.
- Run Jev decision suite for safety, completeness, language/intent confidence and ambiguity.
- Accept at the source-defined confidence gate of 0.85; otherwise clarify, re-run or escalate.
- Validate VideoSpec with Pydantic and persist prompt, revision, model versions and decision evidence.

9.1.5. Workflow



Prompt Intelligence is the first interpretation layer. It accepts free-form natural language and produces a validated VideoSpec containing the information needed by planning and downstream generation modules.

9.1.6. Core Processing

- Language detection and intent classification
- Entity extraction for people, animals, products, objects and locations

- Action and relationship understanding
- Visual extraction: style, lighting, camera, colors and atmosphere
- Technical settings: duration, aspect ratio, resolution and FPS
- Ambiguity detection and clarification questions
- Prompt enhancement without changing user intent
- Negative prompt generation and safety validation
- Pydantic schema validation and structured JSON output

9.1.7. Technology Direction

The production technology stack is **Gemini 3.8 Flash + Jev**.

Gemini 3.8 Flash handles all generative and reasoning tasks (extraction, enhancement, and planning), while Jev acts as the fast, type-safe decision layer for safety, ambiguity, completeness, and final confidence gates.

Earlier research options such as Ollama + Qwen2.5:14b (local) and Claude Sonnet (paid) remain documented as alternatives but are **not** the selected production path.

Example VideoSpec Fields

Group	Example Fields
Identity	original_prompt, language, intent
Entities	subjects, locations, objects
Actions	subject_id, action, details
Visual	style, lighting, camera, atmosphere
Settings	duration_seconds, aspect_ratio, resolution, fps
Safety	negative_prompt, ambiguities, safety_status,confidence

9.1.8. Overview

Goal: understand the user's natural-language prompt and convert it into clear, structured instructions for the rest of the video-generation system.

Main output: a validated VideoSpec (structured JSON) containing every requirement the Video Planning Engine (Member 2) needs.

This is the first and most critical stage of the pipeline: the quality of the whole video depends on how accurately this module understands and structures the request.

9.1.9. Objectives

- Accept natural-language prompts from users.
- Detect language, intent, entities, actions, visual style, camera, lighting and technical settings.
- Detect missing or unclear information and ask clarification questions when required.
- Enhance simple prompts into detailed cinematic descriptions without changing user intent.
- Generate negative prompts and perform safety validation.
- Output a standardized, machine-readable VideoSpec JSON.
- Use the Gemini 3.8 Flash + Jev hybrid for high accuracy, low cost and strong safety.

9.1.10. Requirements & Model Mapping

#	Requirement	Primary model	How it is implemented	Jev role / secondary
1	Prompt input	—	Accept raw free-form text (any language)	—
2	Language detection	Gemini 3.8 Flash (low thinking)	Single structured call returning ISO language code + confidence	Noul: “Is language detection high-confidence?”
3	Intent understanding	Gemini 3.8 Flash	Classify video type: realistic / cinematic / animated / educational / promotional / other + sub-style	Choice for final routing category
4	Entity extraction	Gemini 3.8 Flash	Extract people, animals, products, objects, locations with attributes	Score: entity completeness / ambiguity risk
5	Action understanding	Gemini 3.8 Flash	Map each entity → verb + manner + target	—
6	Visual requirement extraction	Gemini 3.8 Flash	Lighting, color palette, background, style, camera angles, atmosphere	Score: visual complexity
7	Video settings	Gemini 3.8 Flash	Duration, resolution, aspect ratio, fps (with	Choice: apply user value or safe

	extraction		defaults)	default
8	Context understanding	Gemini 3.8 Flash (medium thinking)	Relationships, causality and spatial layout between subjects, objects and actions	—
9	Ambiguity detection	Hybrid – Jev primary	Gemini flags missing / unclear items; Jev runs parallel Noul + Score questions ("Is duration missing?", "Is subject action clear?")	Primary for speed and calibration
10	Clarification system	Gemini 3.8 Flash	Generates natural follow-up questions when essential details are missing	Decides whether clarification is required
11	Prompt enhancement	Gemini 3.8 Flash	Expands sparse prompts into detailed cinematic descriptions while preserving intent (system-prompt controlled)	Score: "Did enhancement stay faithful?"
12	Negative prompt handling	Gemini 3.8 Flash	Explicit "do not include" list	Noul: safety of negative constraints
13	Safety validation	Jev primary	Parallel Noul + Choice questions on policy rules (violence, sexual content, illegal, hate, etc.)	Gemini only for borderline cases needing deeper context
14	Structured JSON output	Gemini 3.8 Flash	Forced structured output → complete VideoSpec schema	Final gate: "Is VideoSpec valid & complete?" (Choice + confidence)
15	LLM research	Decided	Evaluate and select suitable language models – finalized as Gemini 3.8 Flash (generative / reasoning) + Jev (decision layer); see Sections 3.7–3.10 for the supporting market analysis	—

9.1.11. Technology Stack :

Component	Technology	Purpose
Primary intelligence	Gemini 3.8 Flash	Extraction, enhancement, structured JSON
Decision layer	Jev (TypeSafe AI)	Safety, ambiguity, completeness and confidence gates
Validation	Pydantic	Schema validation
Language	Python	Backend implementation
Structured output	Gemini JSON mode + Pydantic	Guaranteed VideoSpec format

9.1.12. VideoSpec Output Format

```
{
  "original_prompt": "A girl walking in the rain at night",
  "language": "en",
  "intent": "cinematic",
  "entities": {
    "subjects": [{
      "id": "subject_1", "type": "person",
      "description": "young woman with long wet dark hair",
      "attributes": ["wearing dark coat"]
    }],
    "locations": ["narrow neon-lit city alley"],
    "objects": []
  },
  "actions": [{
    "subject_id": "subject_1", "action": "walks slowly",
    "details": "looking slightly downward"
  }],
  "visual": {
    "style": "cinematic realistic",
    "lighting": "neon pink and cyan reflections, volumetric rain",
    "camera": { "shot_type": "medium tracking shot",
                "movement": "side follow at shoulder height", "lens": "35mm" },
    "atmosphere": "moody, rainy night"
  },
  "settings": { "duration_seconds": 8, "aspect_ratio": "16:9",
                "resolution": "1080p", "fps": 24 },
  "negative_prompt": ["blurry", "low quality", "watermark", "text",
                      "extra limbs", "deformed face", "flickering"],
  "enhanced_prompt": "A young woman with long wet dark hair ... at night...",
  "ambiguities": [],
  "safety_status": "safe",
  "confidence": 0.94
}
```

9.1.13. Integration with Other Modules

Module	Input from Member 1	How it is used
Member 2 (Video Planning)	Full VideoSpec	Creates VideoBlueprint
Member 3 (Image Generation)	Enhanced prompt + entities + style	Generates keyframes
Member 4 (Video Generation)	Enhanced prompt + camera + motion	Generates video clips
Member 5 (QA)	Original + enhanced prompt	Checks intent match
Member 8 (Backend)	Original prompt + VideoSpec	Project history
Member 10 (Frontend)	Clarification questions	Shows follow-up questions

9.1.14. Service Providers of Models :

The table lists the companies that supply, host or serve the models relevant to prompt understanding, extraction and gating. Selected providers are highlighted.

Provider	Models relevant to Module 1	Access channels	Position in this project
Google	Gemini 3.8 Flash (primary); Gemini 3.5 Flash-Lite; Gemini 3.1 Pro	Gemini API, Google AI Studio; also resold through gateways such as OpenRouter	Selected – generative provider
TypeSafe AI	Jev 1.13 (“System One” typed-decision model)	Early-access API and Python / JavaScript SDKs; also via Vercel AI Gateway, OpenRouter, APIMaster, Requesty and Cloudflare	Selected – decision-layer provider
OpenAI	GPT-6 Sol, GPT-6 Luna (GPT-5.6 tiers still listed); Moderation endpoint	OpenAI API	Alternative / benchmark
Anthropic	Claude Haiku 4.5, Claude Sonnet 5, Claude Opus 5.5	Anthropic API, Amazon Bedrock	Fallback provider
DeepSeek	DeepSeek V4.1 Flash	DeepSeek API	Low-cost alternative
Alibaba Cloud (Qwen)	Qwen3.7 Flash, Qwen3.8-Flash	Alibaba Cloud API (global deployment)	Lowest-cost alternative
Mistral AI	Mistral Small 4	Mistral API	Low-cost EU alternative
Meta (open weights) / self-hosting	Llama Guard family (safety classifiers)	Self-hosted on own GPUs	Alternative safety classifier

9.1.15. Competitors in the Market :

Competing models for the two roles Module 1 needs: (a) generative extraction and enhancement, and (b) fast decision gating.

Layer	Competitor (provider)	Price in / out ($ per 1M)	Strengths	Why it is not the primary choice
Generative	Gemini 3.8 Flash (Google) – selected	$0.75 / $3.75	1M context, tunable thinking, structured output, strong agentic performance	— (chosen)
Generative	GPT-6 Sol (OpenAI)	$2 / $10	Strong general reasoning and structured outputs	About 2.7x Gemini's cost; no measured gain needed for extraction
Generative	Claude Sonnet 5 (Anthropic)	$2 / $10	Strong instruction following and extraction; 1M context	About 2.7x Gemini's cost; same reasoning as above
Generative	Claude Haiku 4.5 (Anthropic)	$1 / $5	Fast; suited to classification and routing; 200K context	Costs more per token than Gemini 3.8 Flash; smaller context
Generative	GPT-6 Luna (OpenAI)	$0.10 / $0.50	Very low cost; high volume	Economy tier; risk of missed details in complex prompts
Generative	Gemini 3.5 Flash-Lite (Google)	$0.30 / $2.50	Low cost; same vendor ecosystem	Lighter reasoning tier; less headroom on nuanced prompts
Generative	DeepSeek V4.1 Flash	$0.15 / $0.60 (off-peak)	Very low cost; 1M context	Data-governance review needed; peak pricing is double
Generative	Qwen3.7 Flash (Alibaba)	$0.03 / $0.13	Cheapest listed API; multilingual strength	Unvalidated on our VideoSpec schema; governance review needed
Generative	Mistral Small 4	$0.15 / $0.60	Low cost; EU-hosted option	Unvalidated on our schema; smaller capability class

Decision	Jev (TypeSafe AI) – selected	$0.042 / free	Typed answers, calibrated probabilities, parallel questions, 70–500 ms	— (chosen)
Decision	LLM-as-classifier (Luna, Haiku, Flash-Lite)	0. 10–1.00 in; paid output	Flexible; also writes free text	Text output must be parsed; no native calibration; slower
Decision	Llama Guard / fine-tuned classifiers	Self-hosted (GPU cost)	Full control; open weights	Safety-only; MLOps burden; no choice/score primitives
Decision	OpenAI Moderation endpoint	Free	Free safety screening	Safety categories only; not a general decision layer

9.1.16. Model Comparison :

Cost per video covers only the Module 1 step, using the average-case token budget: Gemini 3,600 input / 2,000 output tokens; decision layer 900 input tokens (plus 300 output tokens for LLM classifiers).

Layer	Model	Provider	Price in / out ($ per 1M)	Est. cost per video (Module 1)	Assessment for Module 1
Generative	**Gemini 3.8 Flash**	**Google**	**$0.75 / $3.75**	**$0.0102**	**Selected. Best balance of extraction quality, structured output, tunable thinking and price.**
Generative	**Gemini 3.5 Flash-Lite**	**Google**	**$0.3 / $2.5**	**$0.00608**	**Cheaper, lighter tier. Candidate fallback for simple prompts only.**
Generative	**GPT-6 Luna**	**OpenAI**	**$0.1 / $0.5**	**$0.00136**	**Cheaper economy tier, aimed at classification and summarising. A/B-test candidate.**
Generative	**DeepSeek V4.1 Flash**	**DeepSeek**	**$0.15 / $0.6**	**$0.00174**	**Cheaper, strong value. Needs data-governance and availability review first.**
Generative	**Qwen3.7 Flash**	**Alibaba**	**$0.03 / $0.13**	**$0.00037**	**Cheapest listed API. Not yet validated on cinematic-prompt extraction.**
Generative	**Mistral Small 4**	**Mistral AI**	**$0.15 / $0.6**	**$0.00174**	**Cheaper, EU-hosted option. Not yet validated on our schema.**
Generative	**Claude Haiku 4.5**	**Anthropic**	**$1 / $5**	**$0.0136**	**Higher per-token price than Gemini 3.8 Flash. Kept as a fallback provider.**
Generative	**GPT-6 Sol**	**OpenAI**	**$2 / $10**	**$0.0272**	**Premium tier, about 2.7x the cost, with no measured quality need for extraction.**
Generative	**Claude Sonnet 5**	**Anthropic**	**$2 / $10**	**$0.0272**	**Premium tier, about 2.7x the cost, with no measured quality need for extraction.**
Decision	**Jev 1.13**	**TypeSafe AI**	**$0.042 / free**	**$0.000038**	**Selected. Typed Choice / Score / Noul answers with calibrated probabilities; 70–500 ms.**
Decision	**GPT-6 Luna as JSON classifier**	**OpenAI**	**$0.10 / $0.50**	**$0.00024**	**Generates text that must be parsed; no native calibrated probabilities; sequential decoding.**
Decision	**Claude Haiku 4.5 as JSON classifier**	**Anthropic**	**$1 / $5**	**$0.00240**	**Same limitations as above at a higher price. Suitable as a fallback gate.**
Decision	**Llama Guard / self-hosted classifier**	**Meta (open weights)**	**GPU hosting cost**	**Infra-dependent**	**Safety only; needs own GPUs, labelled data and MLOps. Cannot score ambiguity or completeness.**
Decision	**OpenAI Moderation endpoint**	**OpenAI**	**Free**	**$0**	**Safety categories only; cannot do ambiguity, completeness or routing decisions.**

9.1.17. Justification:

Several models in 3.9 cost far less per token than Gemini 3.8 Flash. The table shows what switching would actually save on the Module 1 step.

Model	Input / output ($ per 1M tokens)	Cost per video	Cost per 100,000 videos	Difference vs Gemini 3.8 Flash
Gemini 3.8 Flash (selected)	**$0.75 / $3.75**	**$0.0102**	**$1,020**	**—**
Gemini 3.5 Flash-Lite	**$0.3 / $2.5**	**$0.00608**	**$608**	**saves $412**
GPT-6 Luna	**$0.1 / $0.5**	**$0.00136**	**$136**	**saves $884**
DeepSeek V4.1 Flash	**$0.15 / $0.6**	**$0.00174**	**$174**	**saves $846**
Qwen3.7 Flash	**$0.03 / $0.13**	**$0.00037**	**$37**	**saves $983**
Mistral Small 4	**$0.15 / $0.6**	**$0.00174**	**$174**	**saves $846**
Claude Haiku 4.5	**$1 / $5**	**$0.0136**	**$1,360**	**costs $340 more**
GPT-6 Sol	**$2 / $10**	**$0.0272**	**$2,720**	**costs $1,700 more**
Claude Sonnet 5	**$2 / $10**	**$0.0272**	**$2,720**	**costs $1,700 more**

What the numbers say

Even the cheapest option (Qwen3.7 Flash) saves only about $983 per 100,000 videos on Module 1. That is a small saving compared with the cost of a wrong VideoSpec, because every downstream job (keyframes, video clips, QA, regeneration) inherits the error.

9.1.18. Why cheaper models are not the primary choice

- Errors are amplified downstream. Module 1 is the foundation of the pipeline. A dropped entity, a misread camera instruction or a wrong duration triggers wasted generation jobs and regeneration, which costs far more than the tokens saved.
- Nuanced extraction needs reasoning headroom. Prompts arrive in any language and mix subjects, actions, lighting, camera and settings. Economy tiers (Luna, Flash-Lite, Small models) are built for classification and summarising and carry a higher risk of missed or over-normalised details.
- Structured-output failures cost money too. A schema-breaking response forces a retry, which erodes the price advantage and adds latency.
- Tunable thinking lets us buy depth only where needed. Gemini 3.8 Flash runs at low thinking for pure extraction and medium for context understanding, so the model is not paying for reasoning it does not use.
- Governance and validation effort. Some of the cheapest options (DeepSeek, Qwen) need a data-governance and availability review before prompts can be sent to them, and none has been validated on our VideoSpec schema.
- **The option stays open.** Phase 3 already includes an A/B test of thinking levels; the same harness can trial Flash-Lite or Luna on simple prompts, routed by Jev's complexity score, and adopt them only if VideoSpec pass rate and human ratings hold.

9.1.19. Why not the premium models (GPT-6 Sol, Claude Sonnet 5)

They cost about 2.7x more per token than Gemini 3.8 Flash (roughly $0.0272 vs $0.0102 per video for this step) with no measured quality gain that extraction requires. They remain useful as benchmarks and as escalation or fallback models.

9.1.20. Why Jev as the decision layer

- **Built for this job.** Jev returns Choice, Score and Noul answers with calibrated probabilities, many questions in one parallel call; an LLM classifier returns text that must be parsed and has no native calibration.
- **Speed and cost.** 70–500 ms per call and about $0.042 per 1M input tokens with free output, i.e. roughly $0.000038 per video for Module 1.
- **Enables the confidence gate.** The ≥ 0.85 rule in Figure 2 depends on calibrated probabilities; this is what makes automatic accept / clarify / reject routing dependable.
- **Wider than the free or open options.** Moderation endpoints and Llama Guard cover safety only, while Jev also scores ambiguity, completeness and intent.

> **Known caveats and fallback plan**
>
> **Jev is an early-access product: text-only, a 64K-token limit per request (state plus all questions), and pricing that TypeSafe says it cannot yet prove is unsubsidised. Mitigation: keep questions short and batched, log every probability, and keep a fallback gate (Claude Haiku 4.5 or GPT-6 Luna with a strict JSON schema) that can be switched on if Jev availability or pricing changes.**

Basis: Rates are derived from the execution costs and token volumes supplied in the source table. The supplied INR examples imply ₹96 per USD.

Component	Pipeline Stage	Input Rate ($ / 1M)	Output Rate ($ / 1M)	Input Rate (₹ / 1M)	Output Rate (₹ / 1M)
Gemini API	Draft VideoSpec / Blueprint	$0.75	$3.75	₹72.00	₹360.00
JEV Decision Suite	Early / Post-Gemini Gates	$42.00	Free / 0	₹4,032.00	Free / 0

9.1.21. Execution Cost Validation

Execution	Token Volume	USD Cost	INR Cost
Gemini Call #1	~1,000 input + ~800 output	$0.003750	₹0.36000
Gemini Call #2	~2,000 input + ~2,500 output	$0.010875	₹1.04400
JEV Fast Gate	~300 input + 0 output	$0.0000126	≈₹0.00121
JEV Decision Suite #1	~1,200 input + 0 output	$0.0000504	≈₹0.00484
JEV Decision Suite #2	~1,500 input + 0 output	$0.0000630	≈₹0.00605

9.2 Module 2 — Video Planning & Orchestration

9.2.1 Overview

Goal: decide how the requested video will be generated and organize the complete generation process.

Main output: a machine-readable VideoBlueprint containing segments, motion, timestamps, dependencies and generation jobs.

The module takes the validated VideoSpec from Member 1 and produces an executable plan that the generation engines can follow.

9.2.2 Objectives

•Analyze the validated VideoSpec.

•Decide continuous clip vs multi-segment structure.

•Allocate durations and plan visual content, motion and keyframes.

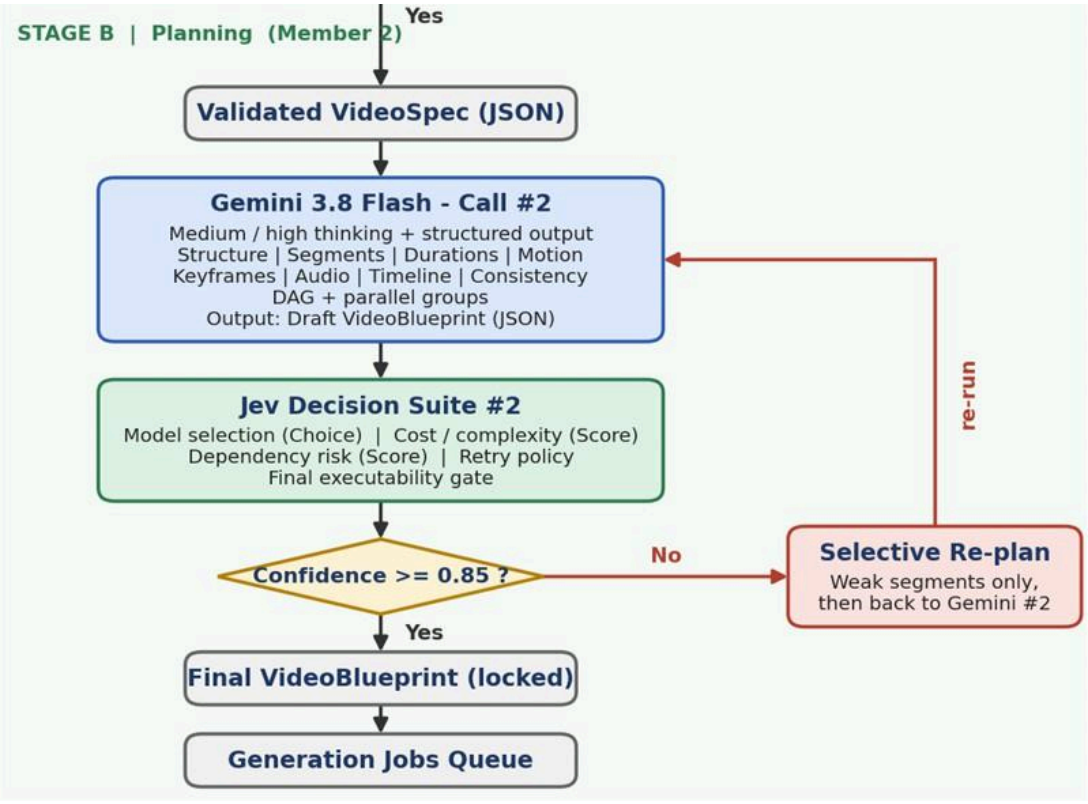
•Build task dependencies (DAG) and identify parallel opportunities.

•Select the most suitable generation models for each task.

•Produce a complete, executable VideoBlueprint.

•Use Gemini 3.8 Flash for planning intelligence and Jev for model selection and final gates.

9.2.3 WorkFlow:



9.2.4. Requirements & Model Mapping

#	Requirement	Primary model	How it is implemented	Jev role / secondary
1	Video requirement analysis	Gemini 3.8 Flash	Reads the full VideoSpec and produces an internal plan summary	—
2	Video structure planning	Gemini 3.8 Flash (medium / high thinking)	Chooses one continuous clip or multiple visual segments	Choice: “single clip or multi-segment?”
3	Segment generation	Gemini 3.8 Flash	Splits into logical visual segments with narrative purpose	—
4	Duration allocation	Gemini 3.8 Flash	Assigns seconds per segment (respects total duration)	Score: duration realism
5	Visual content planning	Gemini 3.8 Flash	Detailed description of what appears in each segment	—
6	Motion planning	Gemini 3.8 Flash	Subject / object / camera movement per segment	Score: motion complexity (affects model choice later)
7	Keyframe planning	Gemini 3.8 Flash	Identifies the critical keyframes to generate first	Choice: mandatory vs optional keyframes
8	Timeline planning	Gemini 3.8 Flash	Absolute start / end times for every visual event	—
9	Consistency planning	Gemini 3.8 Flash	Lists subjects, products and environments that must stay consistent across segments	Noul: “Is consistency list complete?”
10	Model selection logic	Jev primary	Choice per task type (image frames, video model) among Wan / Runway / Veo / Omni / Minimax / etc.	Gemini proposes candidates; Jev makes the final typed decision with confidence
11	Task dependency management	Gemini 3.8 Flash	Builds the dependency graph (DAG)	Score: critical-path risk
12	Parallel processing	Gemini 3.8 Flash + rules	Identifies independent tasks that can run concurrently	Choice: “Can these two jobs run in parallel?”
13	Failure recovery	Jev primary	Retry policy, max retries, backoff, alternative model on failure	Gemini only for complex diagnosis
14	Selective regeneration	Gemini 3.8 Flash + Jev	Gemini identifies which segments need redo; Jev decides scope (single segment vs cascade) – avoids a full restart	Noul: “Safe to regenerate only this segment?”
15	Blueprint generation	Gemini 3.8 Flash	Produces the final machine-readable VideoBlueprint (JSON): segments, motion, timestamps, jobs, dependencies	Final gate: “Is Blueprint executable?”

Queue and retry scheduling (items 12–13) use Jev or classical rules for the high-frequency, low-latency routing decisions; the generation-jobs layer itself is orchestration code, not an LLM, with Jev providing the final pre-job gates (safety, cost, priority).

9.2.5. Technology Stack

Component	Technology	Purpose
Primary planning	Gemini 3.8 Flash	Structure, segments, motion, DAG, blueprint
Decision layer	Jev	Model selection, risk scoring, final gate
Validation	Pydantic	Schema validation
Language	Python	Backend

9.2.6. VideoBlueprint Output Format

```
{
  "video_id": "vid_001",
  "total_duration": 8,
  "structure": "multi_segment",
  "segments": [{
    "segment_id": "seg_01", "start": 0, "end": 4,
    "visual_description": "...",
    "motion": "side tracking shot",
    "keyframes": ["kf_01", "kf_02"],
    "model": "runway_gen3"
  }],
  "dependencies": [{ "from": "kf_01", "to": "seg_01"
  }], "parallel_groups": [["kf_01", "kf_02"]],
  "consistency_list": ["subject_1", "location_alley"],
  "generation_jobs": [ ... ]
}
```

9.2.7. Service Providers of Models

Module 2 needs long-horizon planning with strict JSON output, plus fast routing decisions. The table lists the providers of the planning and decision models, and the video-generation providers that Jev chooses between.

Provider	Models relevant to Module 2	Access channels	Position in this project
Google	Gemini 3.8 Flash (planning); Gemini 3.1 Pro (possible escalation); Veo (video generation target)	Gemini API, Google AI Studio; Google video-generation API for Veo	Selected – planning provider; Veo is a model-selection target
TypeSafe AI	Jev 1.13 (model selection, risk scores, final gate)	Early-access API and SDKs; Vercel AI Gateway, OpenRouter, APIMaster, Requesty, Cloudflare	Selected – decision-layer provider
OpenAI	GPT-6 Sol, GPT-6 Luna	OpenAI API	Alternative / escalation planner
Anthropic	Claude Sonnet 5, Claude Opus 5.5, Claude Haiku 4.5	Anthropic API, Amazon Bedrock	Alternative / escalation planner
DeepSeek	DeepSeek V4.1 Flash	DeepSeek API	Low-cost alternative
Alibaba Cloud	Qwen3.7 Flash; Wan (video generation target)	Alibaba Cloud API	Low-cost alternative; Wan is a model-selection target
Mistral AI	Mistral Small 4	Mistral API	Low-cost EU alternative
Runway, Minimax and other video-model vendors	Runway, Minimax, Omni video models	Vendor APIs	Model-selection targets for Jev (not planners)

9.2.8. Competitors in the Market

Competitors for planning (DAGs, timelines, consistency lists) and for routing and gating decisions.

Layer	Competitor (provider)	Price in / out ($ per 1M)	Strengths	Why it is not the primary choice
Planning	Gemini 3.8 Flash (Google) – selected	$0.75 / $3.75	1M context, medium/high thinking, agentic multi-step planning, strict JSON	— (chosen)
Planning	Claude Opus 5.5 (Anthropic)	$4 / $20	Frontier reasoning for the hardest plans	About 5x Gemini's cost; over-specified for routine planning
Planning	GPT-6 Sol (OpenAI)	$2 / $10	Strong reasoning and structured outputs	About 2.7x Gemini's cost; keep as escalation planner
Planning	Claude Sonnet 5 (Anthropic)	$2 / $10	Strong planning and instruction following; 1M context	About 2.7x Gemini's cost; keep as escalation planner
Planning	Gemini 3.1 Pro (Google)	$2 / $12 (≤ 200K)	Deeper reasoning tier in the same ecosystem	Higher cost; reserve for rare, very complex plans
Planning	Claude Haiku 4.5 (Anthropic)	$1 / $5	Fast; 200K context	Costs more than Gemini 3.8 Flash; smaller context
Planning	GPT-6 Luna / Gemini 3.5 Flash-Lite	$0.10 / $0.50 and $0.30 / $2.50	Low cost	Economy tiers; weaker fit for dependency graphs and consistency
Planning	DeepSeek V4.1 Flash / Qwen3.7 Flash	$0.15 / $0.60 and $0.03 / $0.13	Cheapest listed APIs	Unvalidated on our blueprint schema; governance review needed
Decision	Jev (TypeSafe AI) – selected	$0.042 / free	Typed Choice / Score / Noul, calibrated probabilities, 70–500 ms	— (chosen)
Decision	LLM-as-router (Luna, Haiku, Flash-Lite)	0. 10–1.00 in; paid output	Flexible reasoning about model choice	Parsing needed; no native calibration; slower per decision
Decision	Rules engine / heuristic router	Engineering time	Deterministic, free to run	Cannot weigh motion complexity, cost and risk together; brittle as models change
Decision	Self-hosted fine-tuned classifier	GPU cost	Full control	Needs labelled routing data and MLOps; slow to adapt to new video models

9.2.9. Model Comparison

Cost per video covers only the Module 2 step, using the costed token budget for Stage B: Gemini 2,000 input / 2,500 output tokens; decision layer 1,500 input tokens (Jev Decision Suite #2), plus 300 output tokens for LLM routers.

Layer	Model	Provider	Price in / out ($ per 1M)	Est. cost per video (Module 2)	Assessment for Module 2
Generative	Gemini 3.8 Flash	Google	$0.75 / $3.75	$0.0127	Selected. Agentic multi-step planning, 1M context, medium/high thinking, strict JSON output.
Generative	Gemini 3.5 Flash-Lite	Google	$0.3 / $2.5	$0.00757	Cheaper, lighter tier. Higher risk of weak DAGs and duration errors.

Generative	GPT-6 Luna	OpenAI	$0.1 / $0.5	$0.00169	Cheaper economy tier. Planning depth for DAGs and consistency is unproven.
Generative	DeepSeek V4.1 Flash	DeepSeek	$0.15 / $0.6	$0.00216	Cheaper, strong value. Needs data-governance review; peak/off-peak pricing varies.
Generative	Qwen3.7 Flash	Alibaba	$0.03 / $0.13	$0.00046	Cheapest listed API. Not yet validated on multi-segment planning.
Generative	Mistral Small 4	Mistral AI	$0.15 / $0.6	$0.00216	Cheaper option. Long-horizon planning quality unproven for our schema.
Generative	Claude Haiku 4.5	Anthropic	$1 / $5	$0.0169	Higher per-token price than Gemini 3.8 Flash; 200K context. Fallback only.
Generative	GPT-6 Sol	OpenAI	$2 / $10	$0.0338	Premium tier (about 2.7x cost). Viable escalation model for very complex plans.
Generative	Claude Sonnet 5	Anthropic	$2 / $10	$0.0338	Premium tier (about 2.7x cost). Viable escalation model for very complex plans.
Generative	Claude Opus 5.5	Anthropic	$4 / $20	$0.0676	Frontier tier (about 5x cost). Over-specified for routine planning.
Decision	Jev 1.13	TypeSafe AI	$0.042 / free	$0.000046	Selected. Typed Choice / Score / Noul answers with calibrated probabilities; 70–500 ms.
Decision	GPT-6 Luna as JSON classifier	OpenAI	$0.10 / $0.50	$0.00026	Generates text that must be parsed; no native calibrated probabilities; sequential decoding.
Decision	Claude Haiku 4.5 as JSON classifier	Anthropic	$1 / $5	$0.00260	Same limitations as above at a higher price. Suitable as a fallback gate.
Decision	Rules engine / heuristic router	In-house	Engineering time	$0	Deterministic but brittle; cannot weigh motion, cost and risk together.

9.2.10. Justification:

Module 2 is where planning depth matters most. The table shows what switching to a cheaper planner would save on the Module 2 step.

Model	Input / output ($ per 1M tokens)	Cost per video	Cost per 100,000 videos	Difference vs Gemini 3.8 Flash
Gemini 3.8 Flash (selected)	$0.75 / $3.75	$0.0127	$1,268	—
Gemini 3.5 Flash-Lite	$0.3 / $2.5	$0.00757	$757	saves $511
GPT-6 Luna	$0.1 / $0.5	$0.00169	$169	saves $1,099
DeepSeek V4.1 Flash	$0.15 / $0.6	$0.00216	$216	saves $1,052
Qwen3.7 Flash	$0.03 / $0.13	$0.00046	$46	saves $1,222
Mistral Small 4	$0.15 / $0.6	$0.00216	$216	saves $1,052
Claude Haiku 4.5	$1 / $5	$0.0169	$1,690	costs $422 more
GPT-6 Sol	$2 / $10	$0.0338	$3,380	costs $2,113 more
Claude Sonnet 5	$2 / $10	$0.0338	$3,380	costs $2,113 more

What the numbers say

The cheapest option (Qwen3.7 Flash) would save about $1,222 per 100,000 videos on Module 2. A flawed blueprint (wrong dependency, impossible duration, missing consistency item) wastes image and video generation jobs that cost far more than the planning tokens.

9.2.11. Why cheaper models are not the primary choice

- **Planning is the hardest task in the pipeline.** Splitting a video into segments, allocating durations, building a DAG and listing consistency items is multi-step, long-horizon reasoning. This is where economy tiers are most likely to fail, and where Gemini 3.8 Flash's medium / high thinking is designed to help.
- **Failures cascade.** A bad plan is discovered late, after jobs have been queued. Selective re-planning limits the damage but is not free.
- **Strict schema compliance.** The VideoBlueprint must validate against a Pydantic schema and drive job creation. Malformed output causes retries that erase the saving.
- **Large context headroom.** A 1M-token context window means long or complex VideoSpecs, few-shot examples and cached system prompts fit comfortably.
- **Unvalidated alternatives.** None of the cheaper models has been tested on our blueprint schema; DeepSeek and Qwen additionally need a governance review.
- **The option stays open.** Jev's complexity score can route simple, single-segment videos to a cheaper planner in a later A/B test, while complex plans stay on Gemini 3.8 Flash.

9.2.12. Why not the premium models (Opus 5.5, GPT-6 Sol, Sonnet 5, Gemini 3.1 Pro)

They cost about 2.7x to 5x more per token (for example $0.0338 vs $0.0127 per video for GPT-6 Sol on this step) and no test has shown that routine plans need them. They are the natural escalation tier when Jev's confidence stays below 0.85 after a selective re-plan.

Why Jev as the decision layer

- **Model selection is a typed decision.** Choosing between Wan, Runway, Veo, Omni and Minimax, scoring motion complexity, cost tier and dependency risk, and deciding executability are Choice / Score / Noul questions – Jev's native format.
- **Calibrated probabilities drive selective re-plan.** Figure 5 depends on the ≥ 0.85 gate to decide which segments to redo instead of restarting everything.
- **Negligible cost.** About $0.000046 per video for this step, so the gate can run on every blueprint and every retry.
- **Better than rules or self-hosted classifiers.** Rules cannot weigh several factors together and break as video models change; a fine-tuned classifier needs labelled routing data and MLOps.

Known caveats and fallback plan

Jev is early-access, text-only and limited to 64K tokens per request, so the blueprint must be summarised for scoring rather than sent in full. Keep a rules-based router as a deterministic fallback, and a fallback gate (Claude Haiku 4.5 or GPT-6 Luna with a strict JSON schema) if Jev availability or pricing changes.

9.3 Module 3 — AI Image Generation

9.3.1Production Model Decision

Production decision: GPT-Image-2 is the primary choice for core generation and typography, while FLUX.2 [pro] acts as the secondary or fallback option , Nano Banana 2 is a rapid fallback/iteration option, and Stable Diffusion 3.5/SDXL remains the self-hosted/custom-control option for R&D and controlled workloads.

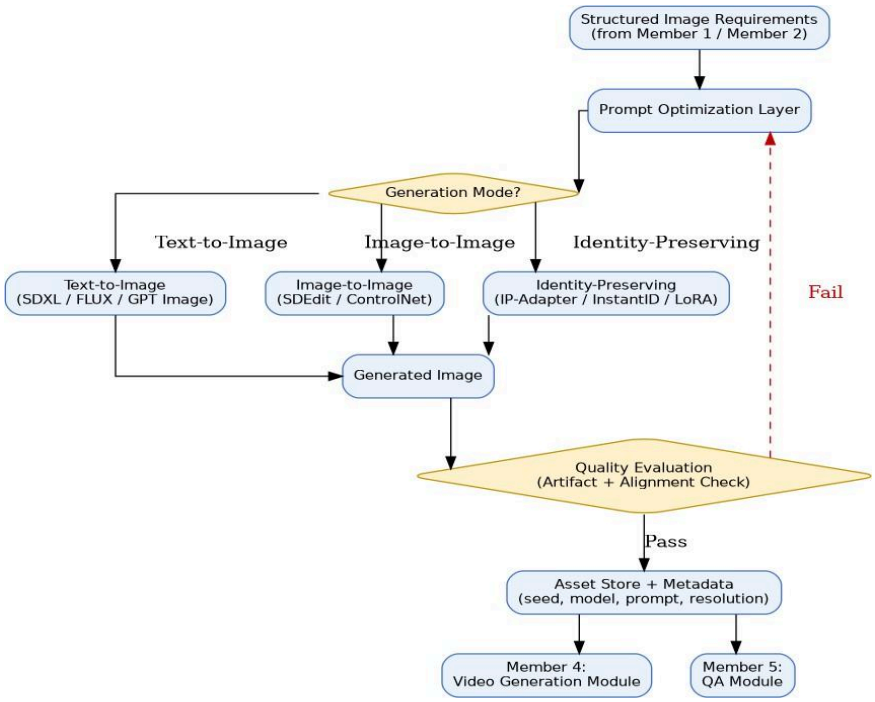
9.3 .2Module Production Model / Technology Comparison

Model	Production role	Why retained	Why not the only model
GPT-Image-2	Primary (Core Generation & Text)	Unmatched prompt adherence, semantic reasoning, and near-perfect typographic text rendering.	Higher per-frame API latency and production generation costs.
FLUX.2 [pro	Secondary / Fast Fallback	Strong multi-reference consistency, structural controls, and high-throughput rendering	Dependent on third-party provider scaling constraints.
Nano Banana 2	Fast fallback	Speed and reference-image workflow.	Not the sole quality/control anchor.
SD 3.5 / SDXL	Self-hosted/custom	LoRA + ControlNet and internal processing.	Team owns GPU, drivers, serving and scaling.
Midjourney	Creative reference	Strong aesthetics.	Source notes no official API in comparison.

9.3 .3 Module Production Implementation Controls

- Create canonical character/product/environment references before keyframes.
- Use multi-reference conditioning, ControlNet and LoRA where supported and legally cleared.
- Record model version, prompt, negative prompt, seed, references and provider request ID.
- Send keyframes through Module 5 QA before using them as video anchors.
- Regenerate only failed assets.

Workflow



The image module functions as the visual foundation or 'casting and set design' stage. It creates references for people, products, animals and environments, and keyframes that anchor video generation.

Core Methods

- Text-to-image diffusion generation.
- Image-to-image editing using controlled denoising.
- Inpainting and outpainting.
- ControlNet for pose/depth/edge structure.
- LoRA for style, character or product adaptation.
- IP-Adapter / InstantID / PhotoMaker / related reference conditioning for identity preservation.
- Seed and generation metadata capture for reproducibility.

Model Landscape in the Source Research

Model / Family	Access	Role in Research
SDXL	Open-source / local	Large ControlNet and LoRA ecosystem
FLUX family	Local or API depending on variant	Prompt adherence, editing/reference workflows
GPT Image	API	High-quality image generation/editing
Midjourney	Web/Discord-oriented	Aesthetic generation; no official API in source
Ideogram	API	Legible text rendering
Adobe Firefly	API / Creative Cloud	Commercial-use-oriented training story

The source module identifies local/open-weight options as cost-efficient for high-volume keyframe generation, while API-based high-quality models can be reserved for selected shots. Exact model availability and pricing should be revalidated at implementation time.

9.4 Module 4 — AI Video Generation & Motion

9.4.1 Production Model Strategy

Use a provider/model adapter architecture rather than hard-coding one video model. The supplied research covers Veo, CogVideoX, LTX, Mochi, HunyuanVideo, Wan and AnimateDiff. Select the first production candidate using project QA, latency, VRAM, licensing, API reliability and cost per accepted second/clip.

9.4.2 Technology Comparison

Family	Strength / role	Why retained	Limitation
Veo	Managed high-quality generation	Useful premium path when quality, provider terms and cost fit.	Managed dependency/per-second cost.
Wan	Open model family	Useful self-hosted cost/control experiments.	VRAM and serving complexity.
HunyuanVideo	High-capability open family	Useful high-quality self-hosted benchmark.	Memory-intensive inference.
LTX	Compact DiT/spatiotemporal	Useful latency/efficiency candidate.	Must be calibrated on project content.
CogVideoX	3D causal VAE + transformer	Useful open-model research.	Hardware/serving complexity.
Mochi	Diffusion transformer	Useful benchmark candidate.	Not universally optimal.
AnimateDiff	Motion modules	Useful controlled motion/custom workflows.	Less general than a full video foundation model.

9.4.3 Selected Wan Family Stack

Wan Model	Task	Input	Production Role	Model Class	Resolution in Selected DPR
Wan 2.2 T2V-A14B	Text-to-Video	Text prompt	Generate scenes directly from prompts	A14B MoE	480p / 720p
Wan 2.2 I2V-A14B	Image-to-Video	Image + prompt	Animate/reference an existing image	A14B MoE	480p / 720p
Wan 2.2 Animate-14 B	Character Animation / Replacement	Character image + motion video	Avatar/character performance	14B	Production character path
Wan VACE	Video Control / Editing	Video + control inputs	Control, transform and refine video	Wan control model	Production editing path

9.4.4 What Each Wan Model Does

Model	Simple Explanation	When We Use It	GPU Intensity	Priority
T2V-A14B	Text → video	User gives only a prompt	■ Very High	Primary
I2V-A14B	Image + text → video	User gives reference image	■ Very High	Primary

Animate-14B	Character image + motion → animated character	Avatar/character scenes	■ Very High	Primary
VACE	Existing video + controls → controlled/edited video	Editing, transformation, control	■ High	Primary

9.4.5. Production Implementation Controls

- Select T2V, I2V or V2V or keyframe-to-video from the VideoBlueprint.
- Route through a stable provider/model adapter.
- Generate short clips with explicit duration, FPS, aspect ratio and motion requirements.
- Send every clip to Module 5 QA before acceptance.
- Retry only for actionable QA reason codes within the retry budget.
- Store model/settings metadata for reproducibility.

9.4.6. Workflow

9.4 Workflow — AI Video Generation & Motion



The video/motion research evaluates text-to-video, image-to-video and keyframe-to-video architectures. Modern systems in the research are primarily latent diffusion transformers operating on compressed spatiotemporal representations, often using 3D VAE components and flow-matching or related objectives.

9.4.7. Canonical Generation Path

Text prompt → text encoder → noise / conditioned latent → spatiotemporal transformer → iterative denoising / flow-matching steps → VAE decoding → video frames.

9.4.8. Generation Modes

Mode	Input	Primary Use
Text-to-video	Text prompt	Generate a clip from a complete description
Image-to-video	Reference image + prompt	Animate an approved keyframe
Keyframe-to-video	Start/end or related keyframes + motion specification	Control continuity and shot evolution
Motion-conditioned	Prompt + pose/trajectory/control signal	Guide subject or camera movement

9.4.9. Research Model Families

- Veo family: commercial latent diffusion transformer approach.
- CogVideoX: 3D causal VAE and transformer architecture.
- LTX family: DiT-based architecture with compact spatiotemporal representation.
- Mochi: asymmetric diffusion transformer approach.

- HunyuanVideo family: high-parameter DiT with memory-intensive inference requirements.
- Wan family: open model family spanning multiple parameter scales.
- AnimateDiff: motion-module approach built around existing image diffusion checkpoints rather than a complete standalone video foundation model.

The research explicitly cautions that generative video models do not have classifier-style accuracy. Evaluation should therefore use project-specific QA plus benchmark dimensions such as VBench, rather than treating one leaderboard or score as definitive.

9.5 Module 5 — Visual Consistency and Quality Assurance

9.5.1 Production QA Decision

Keep QA as a self-hosted, layered diagnostic service rather than a single quality score. The supplied Module 5 research recommends AdaFace as the first face-identity candidate subject to exact checkpoint clearance, DINOv3 for broad visual appearance, RAFT as the initial optical-flow baseline, CLIP for lightweight semantic alignment, LPIPS/SSIM for perceptual/structural checks, selective artifact inspection, and VBench as a benchmark layer.

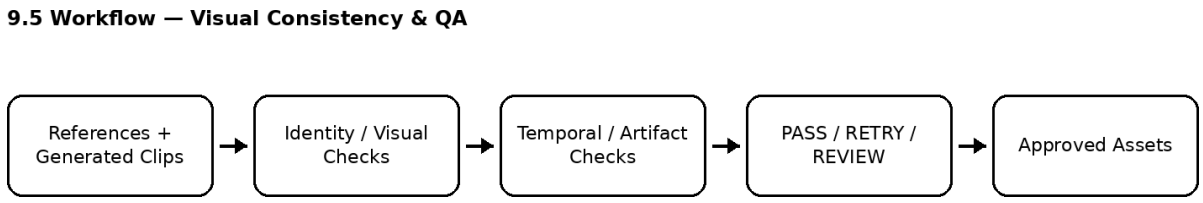
9.4 .2 Module 5 Production Model / Technology Comparison

Dimension	Primary	Why	Alternative / limitation
Face identity	AdaFace candidate	Quality-aware embeddings for identity comparison.	MagFace alternative; exact checkpoint clearance required.
Appearance	DINOv3	Broad visual similarity beyond faces.	Not enough for critical identity/text.
Motion	RAFT baseline	Optical-flow evidence for temporal anomalies.	UniMatch alternative.
Semantic	CLIP	Lightweight first-pass prompt alignment.	VLM escalation for harder cases.
Perceptual	LPIPS + SSIM	Perceptual/structural change detection.	Neither proves semantic correctness.
Benchmark	VBench family	Standardized model/video comparison.	Not a diagnostic production gate.

9.4 .3 Module 5 Production Implementation Controls

- Checkpoint 1: keyframe QA before video generation.
- Checkpoint 2: per-clip identity, temporal, motion, semantic and artifact QA.
- Checkpoint 3: post-interpolation/upscaling QA.
- Use hard failures, calibrated soft scores, AUTO_RETRY and HUMAN_REVIEW.
- Coarse-sample long clips first, localize anomalies, then inspect suspicious ranges densely.

9.5.4 Workflow



QA is the quality gate rather than another generation stage. The module evaluates approved references, generated clips and post-enhancement outputs.

Dimension	Evidence / Method	Decision Use
Subject identity	ArcFace/face embeddings; DINOv3 secondary	Identity drift
Product	DINOv3 + object detection/OCR where needed	Feature/design change
Environment	DINOv3 + CLIP + representative frames	Scene/semantic change
Frame	LPIPS + SSIM + frame difference	Per-frame stability
Segment	Identity/visual embeddings + boundary color checks	Cross-clip continuity
Color	LAB histogram + CIEDE2000	Color outliers

Motion	RAFT optical flow with context	Motion anomaly
Flicker	Frame differences + motion context	Temporal flicker
Artifacts	VLM + detectors/rules	Malformed content
Prompt alignment	CLIP + supplementary VLM	Missing/mismatched requirements

9.5.5 Decision Engine

- HARD FAILURE: severe identity failure, missing required subject or major artifact can force retry/review.
- SOFT SCORE: contributes to composite evidence.
- LOW CONFIDENCE: route to HUMAN_REVIEW instead of forcing an automatic decision.
- RETRY CAP: prevents infinite regeneration and uncontrolled cost.
- POST-ENHANCEMENT QA: revalidate after interpolation or upscaling.

For a 120-frame example, the source proposes coarse sampling every fourth frame, anomaly localization, dense sampling around suspicious ranges and targeted review. This keeps expensive semantic inspection focused where evidence suggests a problem.

9.5.6 Video Accountability, Ownership and Traceability

Accountability answers a different question from QA: which approved asset was created, who is authorized to receive it, what provenance was recorded, and whether a detectable copy appears outside the authorized distribution path.

Recommended flow: Approved Module 5 asset → stable video_id (UUID) → C2PA provenance → recipient-specific VideoSeal watermark → private encrypted storage / authenticated delivery → permitted-source monitoring → pHash candidate retrieval + DINOv3 verification → watermark detection → accountability dashboard → human investigation.

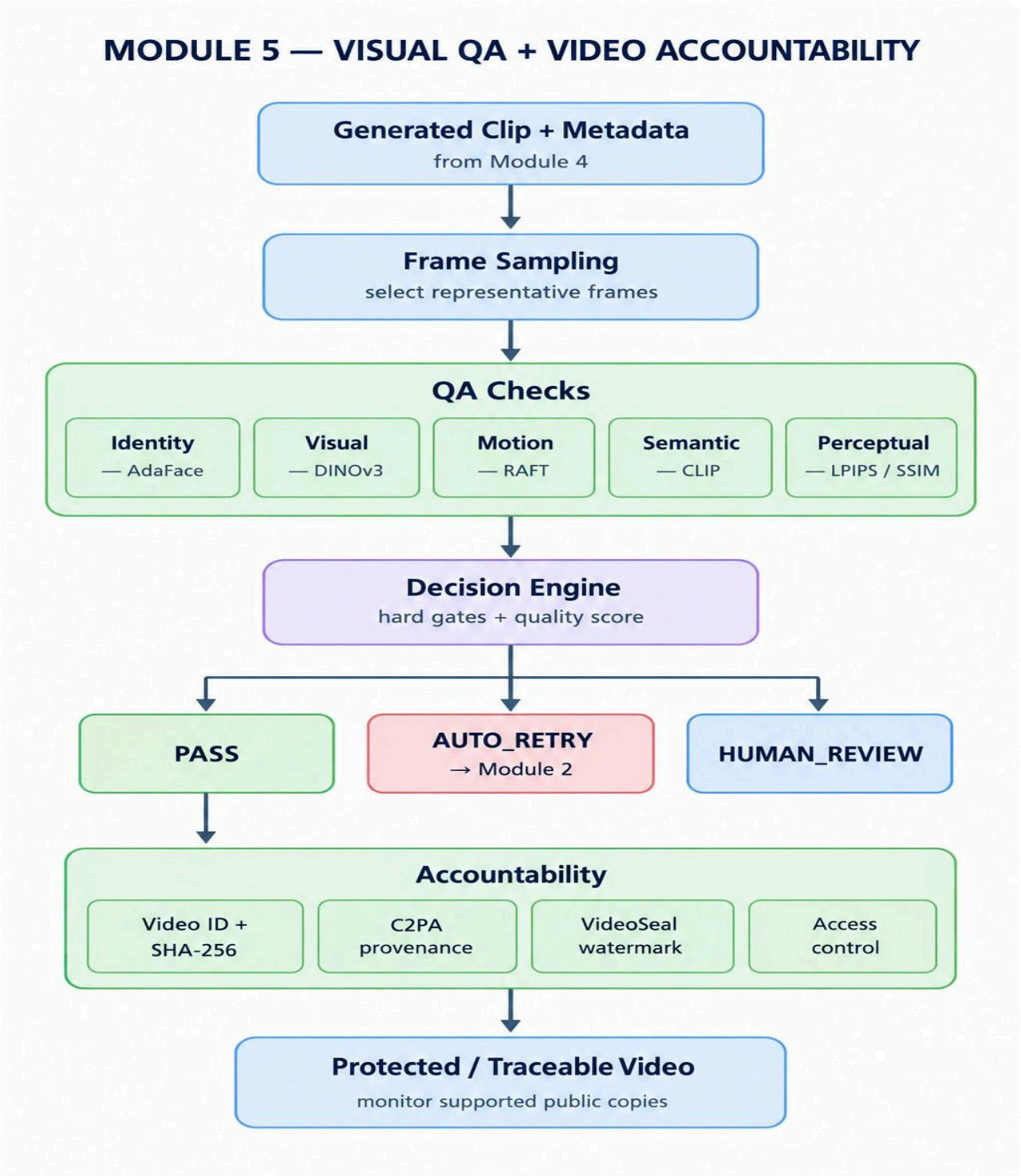
What can be guaranteed: signed origin records and controlled access can be implemented.

What is conditional: tracing a distributed copy through a recovered recipient watermark and detecting modified public copies.

What cannot be guaranteed: universal Internet tracking, private/offline copies, or immunity to every transformation.

Technology	**Production role**	**Why it is used**	**Limitation**
UUID + SHA-256	Stable identity and exact-file integrity	Simple,deterministic ownership record and artifact verification.	Exact-file identity only; edited/compressed copies need other evidence.
Ed25519	SignedOwnership/distribution record	Protects integrity of audit records without raw storing user data in the asset.	Does not itself detect copied media.
C2PA	Provenance / Content Credentials	Records verifiable creation and processing history.	Provenance is not the same as leak detection.
VideoSeal v1.0	Forensic watermark candidate	Embeds recipient-specific hidden token in the media for detectable-copy investigation.	Robustness must be measured under project transformations; not universal protection.
pHash + DINOv3	Copy search / verification	Cheap candidate retrieval first, richer visual verification second.	Only finds copies from supported/permitted sources and can require human review.
Private storage + object access control	Controlled delivery	Keeps masters protected and access auditable.	Cannot prevent screen recording or private/offline redistribution.

Security controls: keep raw customer identifiers out of the watermark payload; store token-to-recipient mapping only in the protected database; protect signing keys; log watermark creation, access, detection and investigation events; monitor only permitted/publicly accessible sources.



9.6 Module 6 — Timeline Synchronization & Rendering

9.6.1 Production Rendering Decision

Use Python for orchestration/timeline logic, GStreamer as the primary modular media framework, ffprobe for pre/post inspection and FFmpeg as the supporting compatibility/utility layer. Use AV1/NVENC AV1 where supported, with H.264/AAC compatibility output. BasicVSR++ is for video super-resolution.

9.6.2 Module 6 Technology Selection and Decision Criteria

Technology	Role	Why selected	Limitation
GStreamer	Primary media framework	Modular element-based pipeline and composition control.	Requires plugin/pipeline engineering.
FFmpeg	Supporting renderer/utility	Mature codec/filter/muxing ecosystem.	Used as utility rather than the only abstraction.
ffprobe	Inspection	Machine-readable input/output validation.	Not a renderer.
AV1	Primary modern codec	Efficient compression for supported targets.	Requires compatibility fallback.
NVENC AV1/B200	GPU encoder	Reduces CPU render time on supported NVIDIA hardware.	Hardware-dependent.
SVT-AV1	CPU encoder	CPU path when GPU encoding unavailable.	Higher CPU time.

9.6.3 Processing Sequence

Stage	Main Action
1. Input	Receive VideoBlueprint and verify mandatory video assets.
2. Preflight	Use ffprobe to inspect media properties.
3. Normalize	Convert incompatible source properties to project targets.
4. Timeline	Create ordered canonical video events and validate timing.
5. Enhance	Apply BasicVSR++; use Real-ESRGAN if the primary model fails.
6. Compose	Trim clips and apply deterministic video transitions.
7. Render	Run GStreamer; use FFmpeg if the primary render fails.
8. Encode	Use NVENC AV1 when supported; otherwise SVT-AV1.
9. Package	Create WebM or MP4 video-only output.
10. Validate	Run post-render ffprobe/QA checks and write the manifest.

9.6.4 Production Implementation Controls

The rendering module uses the following production controls:

- Use a single canonical timeline for accurate video timing.
- Run ffprobe before rendering to validate input videos.
- Normalize FPS, resolution and video format when required.
- Use GStreamer as the primary rendering engine.
- Use FFmpeg as the fallback if GStreamer fails.
- Use BasicVSR++ for video super-resolution and Real-ESRGAN as fallback.
- Use NVENC AV1 for supported GPU encoding and SVT-AV1 for CPU encoding.
- Perform post-render validation for duration, resolution, FPS, codec and file readability.
- Store the render status and fallback information in the render manifest.

9.6.5 Workflow



The workflow begins with the VideoBlueprint and approved video assets. The assets are inspected and normalized before being placed on the project timeline. Video enhancement is applied where required, followed by deterministic composition and rendering.GStreamer is used as the primary rendering engine. If the primary pipeline fails, the same validated composition is re-rendered using FFmpeg. After encoding, the output is validated before being handed to the backend and storage layer.

9.6.6 Fallback and Validation Rules

If GStreamer cannot build the pipeline, exits with an error or times out, the same validated composition is rendered through FFmpeg.

If BasicVSR++ fails because of runtime error, unsupported input, memory failure or timeout, the affected clip is processed with Real-ESRGAN.

Fallback output must preserve the target resolution, FPS and duration.

Duration mismatches are flagged; the system must not silently truncate required content.

Pre-render and post-render checks verify file readability, duration, codec, container, resolution and FPS.

Fallback reasons and final render status are recorded in the render manifest.

9.7 Module 7 — Backend, Database and APIs

9.7.1 Production Backend Decision

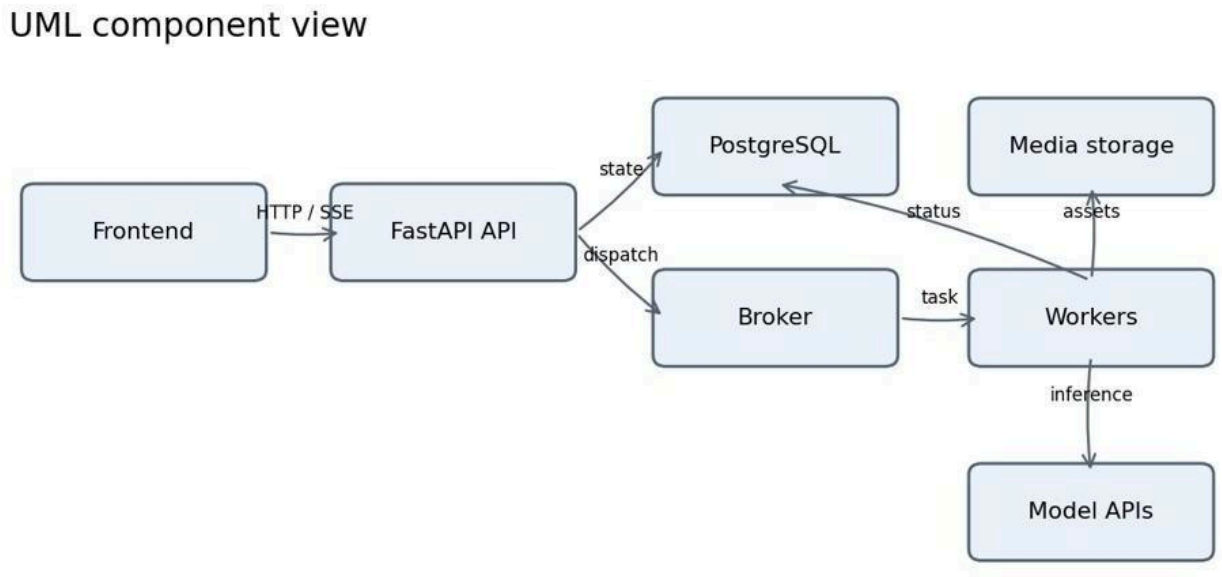
Use FastAPI as the browser-facing API and modular monolith, PostgreSQL as the transactional source of truth, Redis + Celery for asynchronous execution, private S3-compatible object storage for media, and separate CPU/GPU/render workers. Large media bytes should remain outside PostgreSQL.

9.7.2 Production Model / Production Implementation Controls

Component	Selected	Why	Alternative / trade-off
API	FastAPI	Typed contracts, async support and OpenAPI.	Django REST/Flask are viable alternatives.
Database	PostgreSQL–	Transactional state, FKs, indexes and versioned records.	No second primary DB required for MVP.
Queue	Redis + Celery	Long-running jobs, retries and worker pools.	Synchronous HTTP is unsuitable for GPU jobs.
Media	Private S3-compatible storage	Scales large media independently.	Local filesystem only for development.
Workers	Separate CPU/GPU/render pools	Independent scaling and resource isolation.	Single pool creates contention.
Contracts	Versioned REST/OpenAPI	Simple browser integration and explicit schemas.	GraphQL/gRPC only when measured need exists.

- Persist job state before dispatch.
- Use outbox/recovery logic to close transaction-to-task gaps.
- Make worker writes and provider callbacks idempotent.
- Keep PostgreSQL for metadata/state and object storage for media bytes.
- Return HTTP 202 for long-running jobs and expose progress.
- Enforce auth, ownership, rate limits and idempotency keys.

9.7.3 Workflow



The backend research proposes a modular monolith with a browser-facing API, service layer, PostgreSQL, Redis/Celery and private S3-compatible object storage. Separate CPU/GPU workers perform generation and FFmpeg rendering.

9.7.4 Core API Surface

Area	Illustrative Endpoint / Contract
Projects	/v1/projects
Prompts	/v1/prompts
Blueprints	/v1/blueprints
Jobs	/v1/jobs
Assets	/v1/assets
Start generation	POST /v1/projects/{id}/jobs
Progress	GET /v1/jobs/{job_id}

9.7.5 Persistence Rule

- PostgreSQL stores projects, scenes, shots, versions, jobs, asset metadata, QA results and state transitions.
- Object storage stores images, clips, previews, final exports and selected intermediates.
- Redis/Celery stores or coordinates asynchronous task execution.
- An outbox or equivalent mechanism is needed to close the transaction-to-task-publish gap.
- Workers and provider callbacks must be idempotent because duplicate delivery can occur.

9.7.6 Authentication and Ownership

The source recommends password hashing, revocable sessions or short-lived access tokens, and ownership enforcement on every project read/write. Cross-account access, token rotation, logout and rate-limit behavior should be tested explicitly.

9.8 Module 8 — Infrastructure, Deployment and Cost

9.8.1 Production Infrastructure Decision

Adopt a hybrid deployment. Keep prompt/planning workloads on CPU or managed LLM services; use self-hosted GPU capacity for lightweight/custom image and QA workloads where economical; use high-VRAM cloud GPUs or managed video APIs for large video models; use dedicated render workers; and scale worker pools independently.

9.8.2 Production Model / Technology Comparison

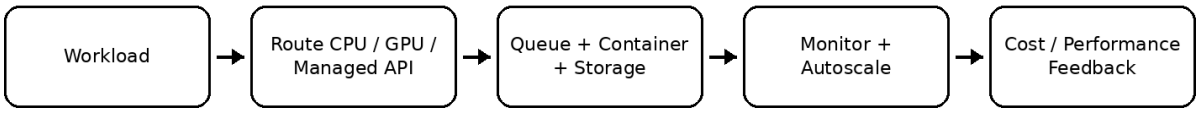
Workload	Recommended	Reason	Alternative / trade-off
Prompt/planning	CPU / managed LLM	Low GPU requirement and easy scaling.	Fully local LLM adds VRAM/maintenance.
Image	Self-hosted or API by model	Balances control, privacy, cost and automation.	All-API raises marginal cost; all-local raises operations.
Large video	80GB+B200 GPU or managed API	Large models are VRAM intensive.	Small GPUs can OOM/model-swap.
QA	B200 class GPU	Good fit for embedding/flow workloads.	VLM escalation can be separate.
Rendering	Dedicated worker	Prevents render contention.	Shared workers contend for resources.
Storage	Private S3-compatible	Scalable and supports lifecycle policies.	Local disk is not production storage.

9.8.3 Production Implementation Controls

- Containerize API, workers and rendering; use CUDA runtime for GPU workers.
- Maintain separate CPU, GPU and render deployments.
- Monitor VRAM, queue wait, model load, failures, retries and cost per accepted output.
- Autoscale only after measuring concurrency and cold-start behavior.
- Version provider/model adapters and revalidate pricing, quotas and licenses.
- Use lifecycle policies for previews/intermediates while preserving required audit evidence.

9.8 Workflow

9.9 Workflow — Infrastructure, Deployment & Cost



Infrastructure is designed around the fact that video generation is memory- and compute-intensive. GPU VRAM is a first-order constraint, while queueing, storage, cold starts, model loading and rendering affect end-to-end latency.

Tier	CPU	RAM	GPU / VRAM	Primary Use
Developer	8–16 cores	32–64 GB	16–32 GB	Prompting, lightweight models, API/UI integration
MVP	16–32 cores	64–128 GB	48–80 GB	Controlled beta and model benchmarking
Production	32–64+ cores	128–512 GB	80 GB+ / multiple GPUs	Concurrent generation
Large-scale	64+ cores	512 GB+	141–180+ GB / multi-GPU	Large models / high concurrency

9.8.4 Hybrid Routing

- Prompt analysis/planning → CPU or managed LLM service.
- Lightweight image generation → self-hosted GPU where practical.
- Large video generation → 80 GB+ B200 GPU or managed video API depending on model.
- Draft preview → lower-cost GPU/model.
- Final rendering → dedicated render worker

B200 Infrastructure

Infrastructure Parameter	Selected Value
AWS Instance	**P6-B200.48xlarge**
GPUs	**8 × NVIDIA B200**
GPU Memory	**180 GB / GPU**
Total GPU Memory	**~1,440 GB**
Planning Server Rate	**~$98.84/hour**
Equivalent Full-Server Rate	**~$1.65/minute**

Infrastructure Parameter	Selected Value
Target Output	**60-sec, 720p**
Target Generation Latency	**60–90 sec**

AWS currently lists the P6-B200.48xlarge at **$98.84/hour** in its Capacity Blocks pricing table for the listed regions, including Mumbai and Hyderabad.

GPU cost required for the models

Pricing Category	Technology	Runtime Allocation	Low	Central	High
1. WAN AI Generation	T2V/I2V + Animate + VACE	60–90 sec	$2.00	**$2.50**	$3.00
2. Rendering	GStreamer + FFmpeg	5–10 sec equivalent	$0.15	**$0.25**	$0.30
3. QA	AdaFace + DINOv3 + RAFT + CLIP + LPIPS/SSIM + VBench	5–30 sec equivalent	$0.15	**$0.50**	$0.82
TOTAL	All three	—	**$2.30**	**$3.25**	**$4.12**

10. Overall Project Flowchart

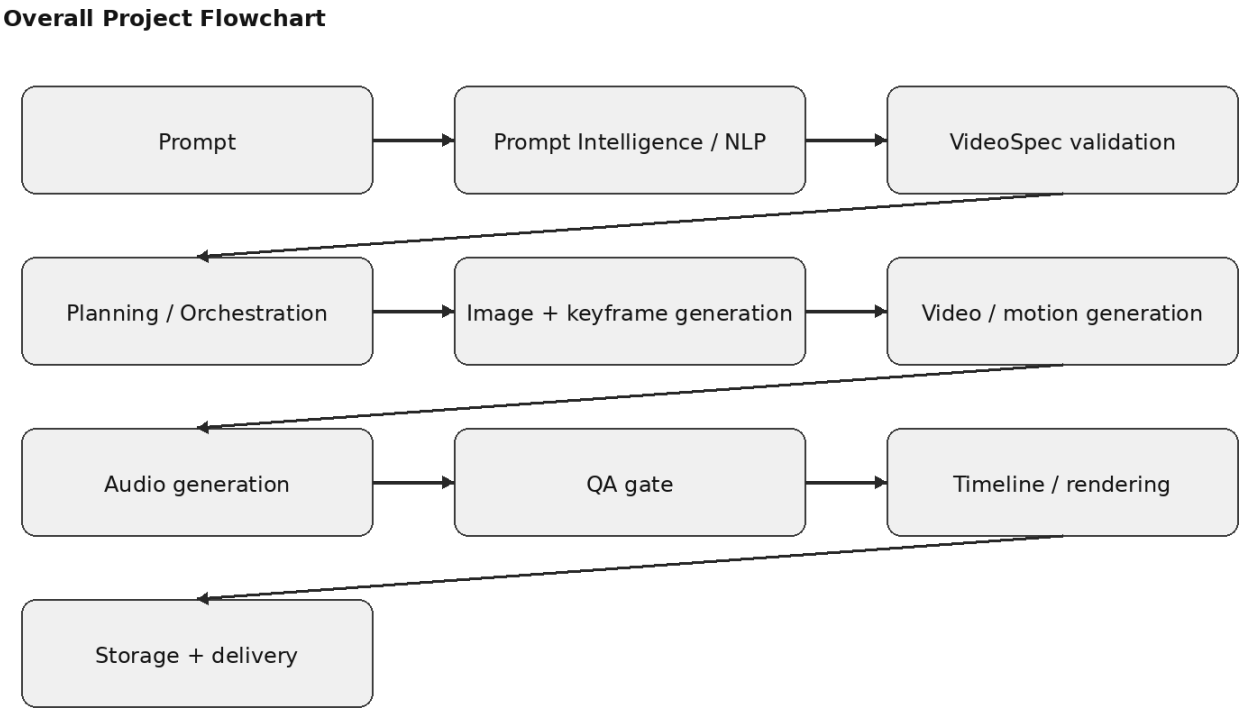


Figure 2. Consolidated end-to-end workflow

The flow deliberately places rights/compliance checks before expensive generation, planning before generation, QA before final rendering, and persistence throughout the lifecycle. Regeneration is treated as a controlled branch. For synthetic content covered by applicable IT Rules, labelling/provenance controls are applied before delivery.

Key Control Points

- Prompt gate: safety, ambiguity and prohibited-use screening.
- Rights gate: uploaded face authorization, consent record and intended-use validation.
- Privacy gate: DPDP purpose, notice/consent, retention, deletion and processor/vendor routing. Blueprint gate: validated production plan before expensive generation.
- Blueprint review gate: user/system review before generation.
- Reference gate: approved visual anchors.
- QA gate: PASS / AUTO_RETRY / HUMAN_REVIEW.

- Render gate: preflight and post-render validation.
- Delivery gate: ownership check and controlled media access.

Synthetic-media delivery gate: apply required user-facing synthetic-content labelling and technically feasible provenance metadata/identifier controls before publication or share-link delivery, subject to the platform's final legal classification.

11. Data Flow Diagram

Data Flow Diagram — End-to-End Generation

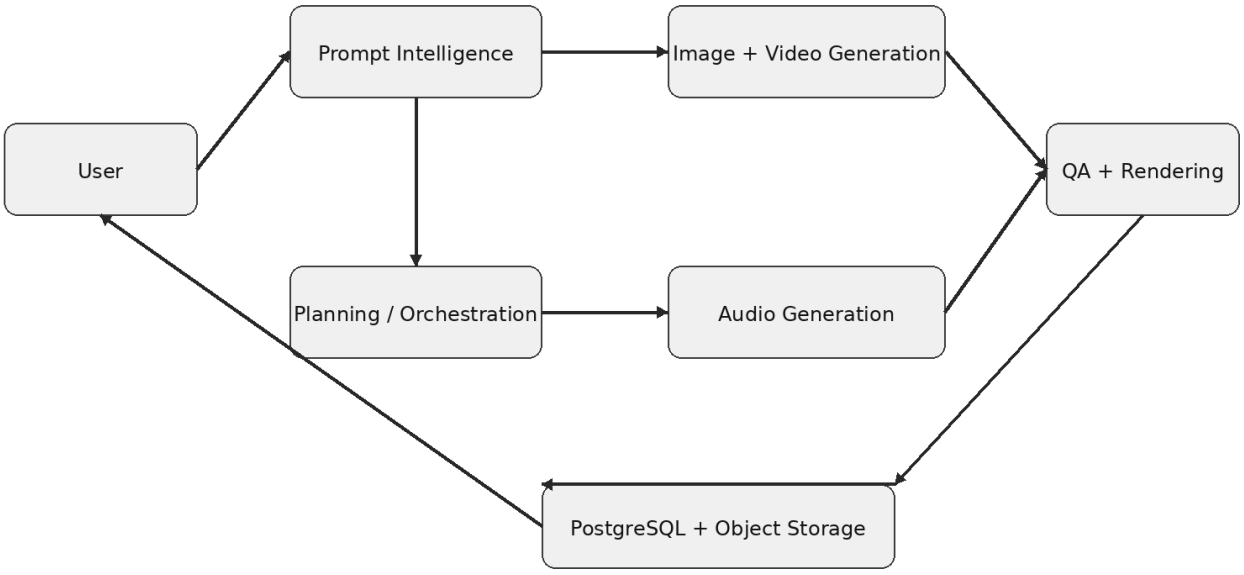


Figure 3. Consolidated data flow

Data Movement Summary

- Prompt data moves from frontend to the API and Prompt Intelligence service.
- Validated VideoSpec moves into planning and becomes a VideoBlueprint.
- Generation workers create media assets and write asset metadata.
- QA consumes references, clips and generation metadata and writes QA evidence.
- Rendering consumes only approved assets and writes the final export and manifest.
- PostgreSQL stores state and relationships; object storage stores large media.
- The frontend receives status and controlled links rather than direct unrestricted storage access.

12. AI/ML Components and Model Strategy

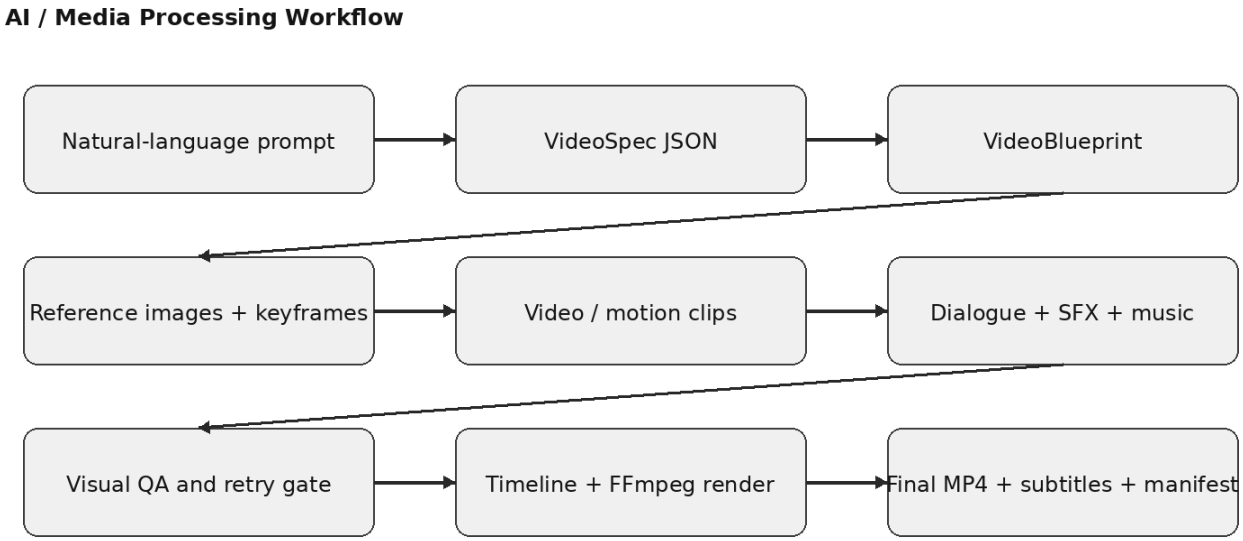


Figure 4. AI/media processing workflow

Function	Candidate / Evidence from Research	Role
Language understanding	Gemini 3.8 flash + jev,Qwen2.5 via Ollama; Claude Sonnet option	Prompt parsing/enhancement
Image generation	SDXL, FLUX, GPT Image, Ideogram, Midjourney	References/keyframes
Identity conditioning	IP-Adapter, InstantID, PhotoMaker, LoRA	Character/product consistency
Video generation	Veo, Kling, Wan, HunyuanVideo, LTX, CogVideoX, Mochi	Motion clips
Visual QA	ArcFace, DINOv3, LPIPS, SSIM, CLIP, RAFT, VLM	Consistency and artifact detection
Interpolation	RIFE / FILM	Intermediate frames
Upscaling	Real-ESRGAN / video-aware upscaler	Resolution enhancement
Rendering	FFmpeg / ffprobe	Composition, encoding and validation

Model choice should remain adapter-based. The research spans rapidly changing commercial and open-weight models; therefore the system should avoid embedding provider-specific assumptions in core business logic.

13. User Workflow

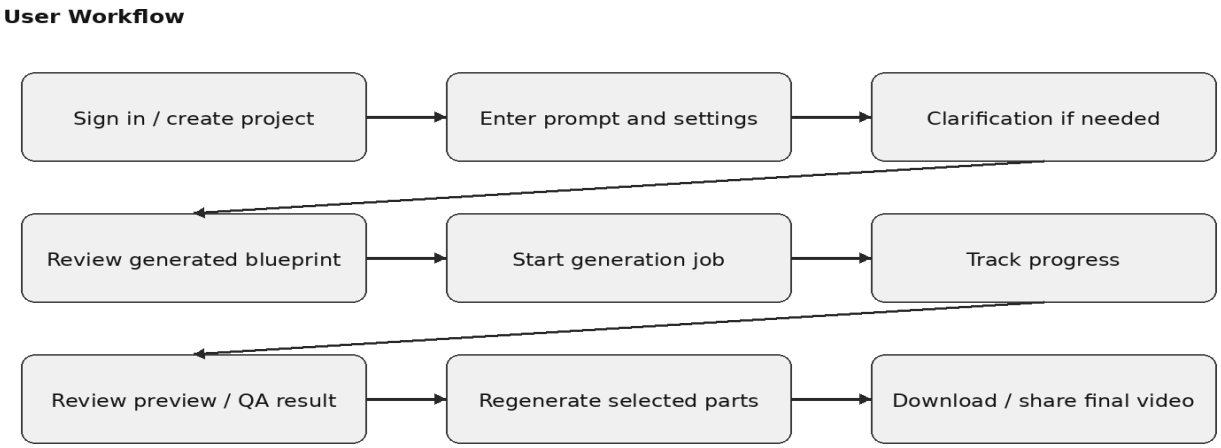


Figure 5. User-facing workflow

User Roles

Role	Responsibilities	Access
User	Create projects, submit prompts, review blueprints, monitor jobs, and review outputs	Own projects/assets
Admin	Review blueprints/previews, request ,regeneration, and monitor jobs, quotas and failures	Project-scoped + operational controls

Legal / Regulatory sources

13.1 Ministry of Electronics and Information Technology — Digital Personal Data Protection Rules 2025 and Enforcement Timeline for the DPDP Act (official document library, November 2025).

13.2 MeitY — Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Amendment Rules, 2026 and official FAQ (effective 20 February 2026).

13.3 Digital Personal Data Protection Act, 2023 — Schedule of monetary penalties; use category-specific statutory maxima rather than a universal ₹250 crore statement.

13.4 Implementation note: all legal dates, classifications, cross-border positions, vendor terms and exact model licences must be revalidated by counsel before commercial launch.

14. Data Design and Security

14.1 Logical Data Model

Entity	Important Fields / Relationships
User	user_id, credentials/session state, status
Project	project_id, owner_id, name, created_at
PromptRevision	prompt_id, project_id, original_prompt, VideoSpec, version
Blueprint	blueprint_id, project_id, version, segments, settings
Scene / Shot	scene_id/shot_id, blueprint_id, timing, prompt, dependencies
Asset	asset_id, project_id, type, storage_uri, checksum, metadata
GenerationJob	job_id, project_id, type, model, status, retry_count, timestamps
QAReport	job_id/clip_id, scores, flags, decision, reason_codes
RenderManifest	render_id, inputs, settings, command/configuration, output
AuditEvent	actor, action, object, timestamp, outcome

14.1 Legal, Rights, Privacy & AI Governance Framework

This section converts the identified legal/compliance gaps into implementation requirements. It is a design-control framework, not a substitute for legal advice. Final classification of the platform, vendor contracts, rights position and applicable obligations must be confirmed by counsel.

14.1.1 Regulatory Mapping

DPDP Act / Rules: the platform should be designed on the assumption that it may act as a Data Fiduciary for direct users and as a Data Processor where a B2B client determines the purpose and the platform processes data on that client's documented instructions. The contract model must reflect the actual relationship for each service.

DPDP implementation controls must cover notice and consent where applicable, children's data, reasonable security safeguards, breach handling, retention and erasure, processor contracts, data-principal rights handling, records and vendor/sub-processor governance.

Timing note: the notified DPDP framework has phased commencement. Rule 4 / the one-year group is scheduled around November 2026, while the main operational provisions and Rules 3, 5–16, 22 and 23 are in the eighteen-month group. The DPR should use 13 November 2026 and 13 May 2027 as planning dates, while counsel verifies the operative Gazette calculation and any later corrigendum.

Penalty note: the ₹250 crore figure is a maximum for specified breaches, notably the reasonable-security-safeguard category; other categories have different statutory maxima. The DPR must not describe ₹250 crore as a universal penalty 'per violation'.

14.1.2 IT Rules 2026 — Synthetic Content

The IT (Intermediary Guidelines and Digital Media Ethics Code) Amendment Rules, 2026 came into force on 20 February 2026. They introduce a due-diligence framework for synthetically generated information (SGI), including technical measures against unlawful SGI and prominent labelling plus permanent metadata or other technical provenance mechanisms where technically feasible.

The DPR should not hard-code a universal three-hour takedown SLA. The 2026 amendments contain several revised timelines, including a three-hour replacement for a particular thirty-six-hour timeline and other category-specific changes. The applicable clock depends on the obligation, complaint/notice type and platform classification.

Open legal question: determine whether the platform's hosting, public sharing or user-to-user features make it an 'intermediary' for the relevant IT Rules. If so, implement grievance intake, escalation, evidence preservation and takedown

workflows to the applicable category/timeline

14.1.3 Rights, Consent and Likeness

Real-person face uploads: require an explicit rights/authorization declaration and a consent record before face embeddings, identity adapters or related derived assets are created. The record should capture purpose, duration, territory, channels, commercial use, training permission, revocation and deletion requirements.

Uploader authorization: provide a mechanism to establish that the uploader is the person depicted or is authorized by that person. Minors require a dedicated restricted workflow and legally appropriate verifiable consent.

Revocation: deletion must cover the source image and, where applicable, face embeddings, identity adapters/LoRAs, cached intermediates, generated outputs and downstream provider copies to the extent the provider contract and law permit/control them.

Impersonation policy: prohibit non-consensual impersonation and define additional restrictions for public figures, celebrities and other real people whose likeness may create heightened deception or rights risk.

14.1.4 Content Safety and Operations

Prohibited-use policy should cover non-consensual intimate imagery, child sexual abuse material, impersonation, fraud/deception, prohibited public-figure impersonation, election/political deepfakes where prohibited by law or policy, and other unlawful content.

Implement input and output moderation, abuse reporting, evidence retention, human escalation and an incident-response path. Where a legal reporting obligation applies, route incidents to the designated legal/compliance owner.

14.1.5 Contracts and Governance

Required legal documents: Privacy Policy, Terms of Service, Acceptable Use Policy, synthetic-content disclosure, grievance mechanism, enterprise DPA, sub-processor list, security addendum and SLA where applicable.

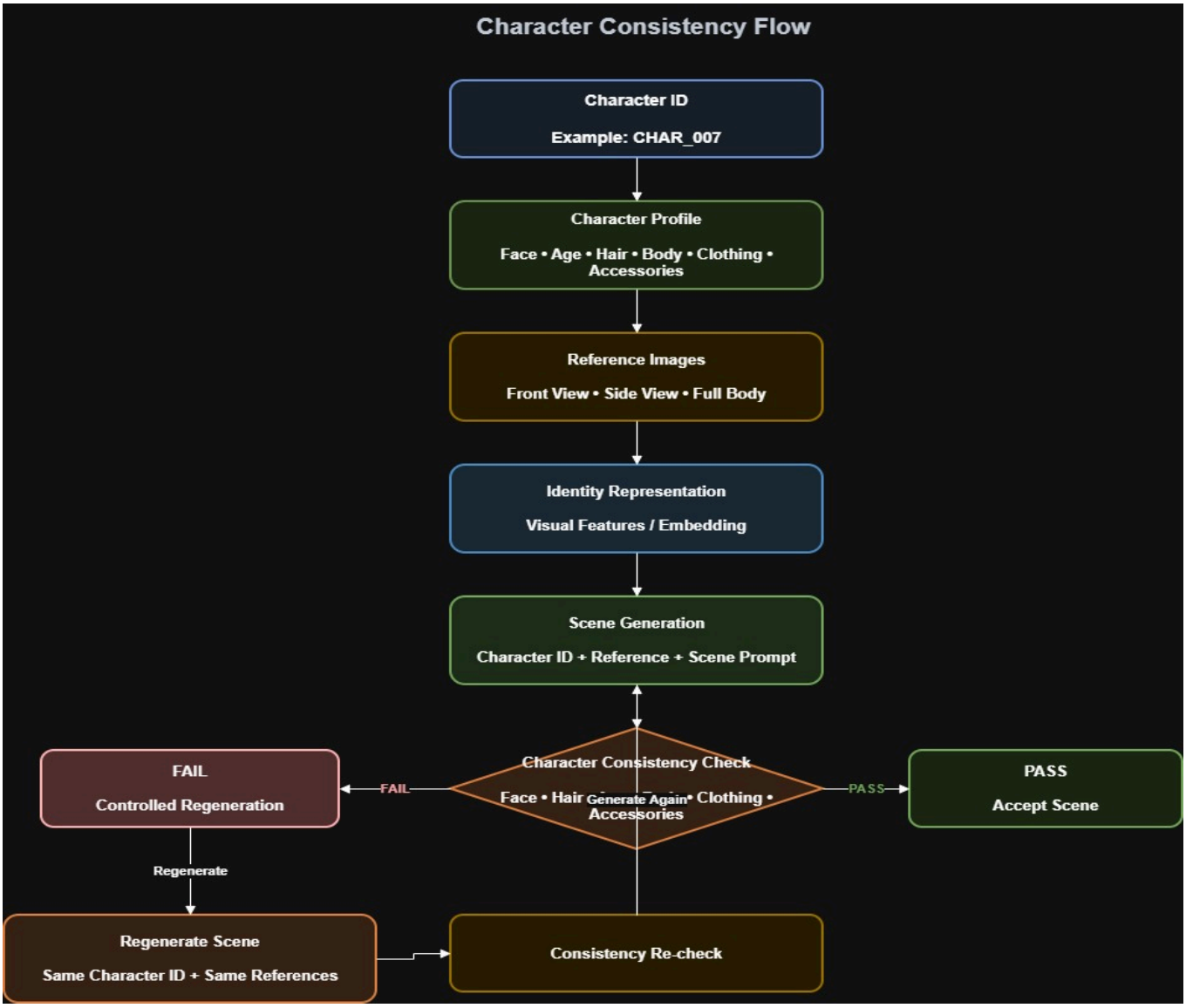
Commercial risk controls: define liability caps, indemnities, insurance requirements, vendor responsibility and customer responsibility.

Governance owner: nominate a privacy/compliance owner (DPO-equivalent or designated contact as legally appropriate) responsible for the compliance register, vendor/model register, rights/consent register, incident workflow and periodic review.

. Compliance Gap Matrix

- Identifies legal, privacy, security, and safety requirements.
- Compares requirements with the current system implementation.
- Identifies missing or partially implemented controls.
- Covers user consent and identity/face rights.
- Checks DPDP and personal-data protection requirements.
- Addresses copyright and content ownership.
- Defines controls for AI-generated/synthetic media.
- Identifies prohibited-content and safety controls.
- Covers data retention and deletion.
- Checks authentication and access control.
- Ensures audit logging and traceability.
- Covers third-party AI/data processors.
- Identifies regional or jurisdiction-specific requirements.
- Defines mitigation actions for each identified gap.
- Tracks the implementation status of each compliance control.
- Provides evidence that compliance requirements are considered during system design and implementation.

14.1.6 Rights, Privacy and Synthetic-Media Control Flow



Compliance Area	Gap Priority	Required DPR / Product Control
DPDP role mapping	High	Document Data Fiduciary vs Processor role per service and contract.
Notice / consent	High	Implement purpose-specific notice, consent and withdrawal where required.
Children	High	Restricted workflow and verifiable consent; no unrestricted face cloning.
Breach / security	High	Incident playbook, logs, notification workflow and vendor escalation.

Retention / erasure	High	Retention schedule and cascading deletion of derived assets.
Face	High	Separate consent records and uploader authorization checks.
Revocation	High	Delete/revoke source and derived assets where applicable.
Vendor DPAs / cross-border	High	Data-flow map, region, retention, training and subprocessors register.
Model/checkpoint licences	High	Exact-version commercial licence register.
Synthetic labelling/provenance	High	Label SGI where applicable and add feasible provenance metadata.
Intermediary classification	High / Open	Counsel to determine hosting/share-link obligations.
Grievance / takedown SOP	Medium–High	Implement category-specific intake and response clocks.
Training-data provenance	High	Dataset licence/consent register; user data training opt-in only.
Output ownership	High / Open	Contract terms plus jurisdiction-specific copyright advice.
Prohibited-use policy	High	Block and escalate prohibited synthetic media.
Audit / provenance records	Medium–High	Retention schedule for manifests and audit events.
Compliance owner	Medium	Nominate accountable owner and review cadence.

14.2 Security Controls

- Hash passwords and use secure session/access-token mechanisms.
- Enforce project ownership on every read and write.
- Keep provider/API secrets outside source code and user-visible payloads.
- Validate uploaded files, paths, formats and media metadata.
- Use argument arrays rather than unsafe shell-string concatenation for FFmpeg execution.
- Use private object storage and controlled download links.
- Record audit events for project, job, asset and export actions.
- Apply quotas/rate limits and protect long-running endpoints from abuse.
- Maintain model/checkpoint license records and commercial-use approval

15. Implementation Methodology

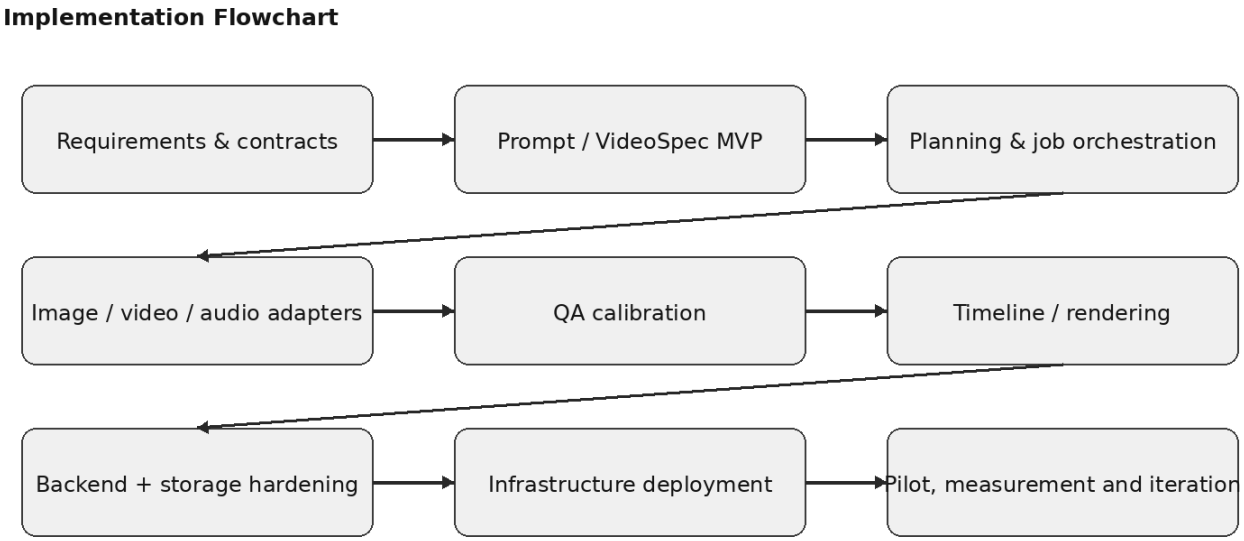


Figure 6. implementation flow

Implementation Strategy

- **Implement user access and project creation** through the login and dashboard, where each project maintains its prompt, character, generated assets.
- **Process the user prompt and establish the character** by allowing either selection of one of the predefined avatars or generation of a new character image. Assign a unique **Character ID** to the selected or generated character and store its reference information.
- **Pass the prompt and Character ID through the JEV layer** to prepare the structured input required by the scene-orchestration stage.
- **Use the Gemini Flash scene orchestrator to convert the creative request into a structured Video Blueprint**, defining scenes, durations, actions, camera instructions, character usage, and generation requirements.
- **Submit the Video Blueprint to the Wan 2.2 video-generation pipeline**, generating scene-level video clips using the associated Character ID and reference information.
- **Perform character tracking and consistency validation** on the generated clips to identify deviations in character identity or appearance. Failed clips are selectively returned to the Wan 2.2 generation stage for regeneration.
- **Assemble the approved scene clips using FFmpeg**, performing trimming, ordering, synchronization, transitions, and final video encoding according to the project configuration.
- **Run the final quality check on the rendered video**. Videos that pass validation are saved and delivered to the user, while failed outputs are routed back to the required generation or processing stage for correction.
- **Measure the complete workflow during pilot deployment**, including generation success rate, regeneration rate, processing time, quality, GPU/resource usage, and cost per accepted video, and use these results to calibrate the production configuration.

15.1 Team Distribution

Ownership follows the module structure of this DPR. Member names are to be assigned by the project lead.

Member	Module Ownership	Primary Work
M.J.R.Kanaka Raju	Prompt Intelligence & NLP (Module 1)	Gemini + Jev pipeline, clarification, VideoSpec schema and validation
K. Vaishnavi	Video Planning & Orchestration (Module 2)	VideoBlueprint, DAG, model selection, retries, selective regeneration
K.Shiva	AI Image Generation (Module 3)	GPT- image-2,FLUX.2 PRO,Reference images, keyframes, identity conditioning, image adapters
A. Srimanth	AI Video Generation & Motion (Module 4)	Wan T2V / I2V / Animate / VACE adapters, provider router, clip metadata
M.Kannan	Visual Consistency, QA & Accountability (Module 5)	QA models, decision engine, calibration set, C2PA and watermark
Jetti Kavya	Timeline Synchronization & Rendering (Module 6)	ffprobe, BasicVSR++, GStreamer / FFmpeg, encoding, render manifest
K. Sai Harshitha	Backend, Database & APIs (Module 7)	FastAPI, PostgreSQL, Redis / Celery, storage, authentication and ownership
C. Kunal	Infrastructure, Deployment & Cost (Module 8)	EKS, GPU routing, monitoring, autoscaling, cost per accepted video
N.Vijay	Frontend & User Workflow	Project, prompt, clarification, blueprint review, progress, preview and download screens
Compliance owner	Legal, Rights, Privacy & AI Governance	Consent and revocation, DPDP, IT Rules labelling, licence and vendor registers, grievance workflow

16. Testing and Quality Assurance

Test Level	Examples	Primary Measures
Prompt	100–200 diverse prompts, incomplete prompts, multilingual inputs	Schema compliance, ambiguity detection, response time
Planning	Dependency and retry scenarios	Correct DAG ordering, state transitions
Generation	Model/API success and failure	Clip success rate, latency, VRAM use
Visual QA	Injected identity/artifact/motion defects	False accept/reject, threshold calibration
Rendering	FPS/resolution mismatches, missing/corrupt assets	Render success, drift, output validity
Backend	401/403/409/429, duplicate tasks	Authorization, idempotency, contract compliance
Load	Fixed concurrent jobs	p50/p95 latency, queue wait, worker utilization
End-to-end	Complete prompt-to-MP4 workflow	Accepted-video rate, cost per accepted output

16.2 QA Metrics

- Identity-score distribution and false-match/false-rejection rates across project-defined appearance groups.
- Prompt alignment, visual consistency, temporal stability and motion evidence.
- Timeline drift in milliseconds and subtitle timing error.
- GPU busy time, queue wait time, cold-start time and peak VRAM.
- Failure/retry rate and cost per accepted output.
- Final media checks using ffprobe for streams, duration, FPS, resolution and encoding metadata.

16.3 Accountability Metrics

- Video identity and ownership record integrity using UUID, SHA-256 and Ed25519 signatures.
- C2PA provenance presence and verification for approved final assets.
- Watermark embedding and recovery success for recipient-specific VideoSeal tokens.
- Candidate-copy detection using pHash/DINOv3, including match accuracy and false-match rate.
- Access-control, watermark and investigation events recorded in audit logs.
- Detection coverage and latency measured only for permitted/publicly accessible sources.

16.4 Source Performance Claims to Treat as Targets

The orchestration research reports blueprint overhead below 1.2 seconds, FFmpeg stitching below 4.5 seconds for a 15-second 1080p video, and a target of under 60 seconds total generation. These are source-reported targets/benchmarks, not results established by this consolidated DPR; they should be measured on the final workload and hardware.

17. Expected Results and Advantages

17.1 Expected Results

- A validated structured representation of each user request.
- An explicit scene/shot execution plan with dependency-aware job scheduling.
- Reference images and keyframes that can be reused across generation.
- Video clips with generation metadata and traceable model settings.
- Layered assets with timing and subtitle metadata.
- Evidence-backed QA decisions and controlled regeneration.
- A synchronized final video with a render manifest and validation record.
- Persistent project history and controlled asset access.

17.2 System Advantages

Advantage	How the Architecture Provides It
Modularity	Each media function is independently replaceable
Consistency	References + identity/visual/temporal QA
Scalability	Async queues and independently scalable workers
Cost control	Routing, previews, coarse QA and retry caps
Reproducibility	Prompt/model/seed/settings/manifest metadata
Security	Auth, ownership, private media and safe execution
Operational visibility	Jobs, logs, metrics and failure reason codes
Extensibility	Provider adapters and versioned contracts

18. Risks, Limitations

Risk / Limitation	Impact	Mitigation / Open Item
Identity drift	Character changes across frames/clips	Reference conditioning + ArcFace/DINOv3 QA; calibrate thresholds
Visual artifacts	Malformed face/hand/object	VLM/detector checks + retry/review
Temporal instability	Flicker, jumps, inconsistent motion	RAFT + frame checks + post-enhancement QA
High VRAM requirement	OOM or expensive infrastructure	Model-specific GPU routing and memory tiers
API timeout/rate limit	Failed generation jobs	Retry/backoff, queues and provider adapters
Retry cost	Unexpected compute spend	Retry caps, confidence routing and previews
Model/provider changes	Broken integrations or changed pricing	Versioned adapters and periodic revalidation
Licensing uncertainty	Commercial deployment risk	Exact checkpoint/provider license review
Long videos	High compute/storage and timeline complexity	Segment rendering, proxies, resumable jobs
Threshold uncertainty	False accepts/rejects	Project-generated calibration set with human labels
Storage growth	Cost and privacy risk	Lifecycle policies and separate intermediate/final storage
Unknown production concurrency	Infrastructure sizing uncertainty	Load-test actual target workload

19. Implementation Roadmap and Future Scope

Phase	Focus	Exit Criteria
Phase 1	Strict Schema, FastAPI Architecture, Redis/Celery Orchestration, Metadata Persistence, Secure S3 Storage, Tenant Security & Controls, Governance Database Fields, System Observability, Secure CLI Execution.	Stable API lifecycle, Decoupled Queue Execution: Redis/Celery, Strict Schema Enforcement, Separated Data Persistence: PostgreSQL, Compliance & Monitoring Readiness.
Phase 2	All Project Done by scratch	A valid, complete, and executable VideoBlueprint/DAG is generated from diverse user prompts, ready for the Image & Video Generation phase

Market Research & Competitive Analysis

19.1 Market Research and Industry Overview

AI-based video generation is developing rapidly as generative AI models become capable of converting text prompts, images and structured instructions into short video sequences. The market is expanding across content creation, advertising, education, entertainment, e-commerce, social media and enterprise communication.

Published market estimates vary because different research companies use different definitions of the AI-video market. Grand View Research estimates the global AI video generator market at approximately USD 788.5 million in 2025 and USD 946.4 million in 2026, with a projected CAGR of 20.3% from 2026 to 2033. The Business Research Company estimates the AI video generator market at approximately USD 1.04 billion in 2026 and projects approximately USD 2.07 billion by 2030, representing an 18.9% CAGR. These figures should therefore be treated as market indicators rather than a single definitive market value.

The broader AI-video market is considerably larger because it includes video analysis, AI-assisted editing and other video applications in addition to generative video. Grand View Research estimates this broader AI-video market at USD 4.6 billion in 2025 and projects USD 42.3 billion by 2033. This distinction is important because the proposed project focuses specifically on an end-to-end AI video generation and production workflow rather than the entire AI-video industry.

19.1.1 Indian Market Context

India represents an important environment for AI-assisted video production because of its large digital audience, regional-language ecosystem, creator economy and expanding online-video consumption.

PwC India projects that India's entertainment and media industry will grow from approximately USD 25.7 billion in 2025 to USD 36.7 billion by 2030, representing a 7.4% CAGR. The report identifies AI, connectivity, digital ecosystems and regional content as important factors shaping the industry's development.

Media Partners Asia data reported in 2026 indicates that online streaming represented approximately 46% of India's video-content investment in 2025, compared with 42% for television. This indicates a continuing shift of video production and investment toward digitally distributed content.

An earlier India-focused video-sector report estimated approximately USD 5.8 billion of screen-content investment in India during 2024 and reported that digital formats such as SVOD, AVOD and user-generated content accounted for approximately 37% of investment, up from 15% in 2019.

These trends create potential demand for tools that can reduce the time and complexity required to plan, generate, evaluate and assemble video content.

19.1.2 **Competitor Comparison**

Platform	Primary Focus	Text-to-Video	Image/Reference-to-Video	Character/Element Consistency	Multi-shot / Story Workflow	API / Developer Integration	Main Market Segment
OpenAI Sora 2	General generative video	Yes	Yes	Character/likeness controls	Story/remix-oriented	Planned/API history; product discontinued Apr. 2026	Creative generation
Google Veo	High-quality generative video	Yes	Yes	Reference-based consistency	Scene/story-oriented controls	Google ecosystem/API access	Filmmakers, creators, enterprises
Runway	Creative AI production	Yes	Yes	Reference/image conditioning	Strong creative workflow	Yes	Creators, studios, enterprises
Kling	Generative video and cinematic control	Yes	Yes	Element consistency	Multi-shot/story board	Platform/API ecosystem	Creators and production

Luma AI	Generative video and modification	Yes	Yes	Reference and visual continuity features	Extension/creative workflow	Yes	Creators, developers
Pika	Consumer creative video	Yes	Yes	Creative transformations	Template/effect-oriented	Yes	Social creators
Adobe Firefly	Generative creative ecosystem	Yes	Yes	Creative asset workflow	Editing + generation ecosystem	Adobe ecosystem	Creative professionals and enterprises
HeyGen	Avatar/business video	Yes	Yes	Strong avatar identity	Presentation workflow	Yes	Business communication
Synthesia	AI presenter/training video	Script-based	Avatar-based	Avatar consistency	Structured presentation workflow	Enterprise integration	Training and enterprise

The comparison shows that existing platforms provide strong individual generation capabilities, but their product approaches differ. Some focus primarily on generation quality, some on creative editing, some on avatar communication, and others on multi-shot storytelling.

19.1.3 India-Centric Opportunity

India has a large and diverse digital-video ecosystem, with strong growth in streaming, regional content and creator-driven production.

The proposed platform can therefore be designed with India-specific requirements such as:

- regional storytelling;
- Indian environments and cultural references;
- character/reference consistency for Indian subjects;
- scalable cloud/API architecture;
- INR-based cost estimation;
- support for regional content workflows;

privacy and consent controls appropriate to the deployment environment.

The platform should not claim superior performance for Indian or regional characters without dedicated benchmarking. A project-specific evaluation dataset should therefore be developed before making such claims.

19.1.4 Competitive Advantage Areas to Validate

The following areas should be treated as project hypotheses to be validated through testing, rather than unsupported claims of superiority:

Area	Proposed Platform Approach	Validation Method
Story planning	VideoSpec + VideoBlueprint	Human evaluation of generated plans

Multi-scene consistency	Persistent references + QA	Character/scene consistency benchmark
Model flexibility	Adapter/model-router architecture	Provider/model replacement test
Quality control	Automated visual and temporal QA	Human-labelled validation dataset
Cost control	Cost-aware routing and retry limits	Cost per accepted video
Production workflow	End-to-end orchestration	Time-to-final-video measurement
Scalability	Queue + GPU worker architecture	Load and concurrency testing
India-focused content	Project-specific regional benchmark	Evaluation using representative Indian scenes/characters
Governance	Rights, consent, provenance and audit controls	Legal and compliance review

19.1.5 Market Research Conclusion

The AI video-generation market is rapidly expanding, but market estimates differ substantially because research organizations use different definitions of AI video and AI video generators. Multiple industry reports nevertheless indicate strong growth through the second half of the decade.

The competitive landscape includes general-purpose video-generation platforms, creative-production systems and avatar-focused business-video platforms. Current products increasingly provide reference images, character consistency, camera control, multi-shot generation, video modification and developer APIs.

The principal opportunity identified for the proposed project is therefore not simply to build another isolated text-to-video model. Instead, the project focuses on creating an end-to-end production orchestration platform that connects planning, generation, consistency management, automated visual QA, rendering and delivery.

The final commercial positioning should be validated through prototype testing, user studies, benchmark datasets, infrastructure measurements, provider pricing and legal/licensing review.

19.2 Future Scope

- More advanced prompt understanding through multimodal inputs, improved intent detection, and ambiguity resolution.
- Enhanced story planning with long-video memory, stronger narrative continuity, and better scene-to-scene consistency.
- Improved character, object, and environment consistency across images and generated video segments.
- Better motion realism, camera control, and temporal coherence using next-generation video generation models.
- Intelligent model routing based on measured quality, cost, latency, and reliability metrics rather than static selection rules.
- More automated image, video, using multimodal AI evaluators and benchmark datasets.
- Interactive scene-level editing and regeneration without rerendering unaffected portions of the final video.
- Advanced multilingual support with automated dubbing, subtitle generation, localization, and stronger lip synchronization.
- Enhanced production capabilities through autoscaling, multi-GPU scheduling, real-time generation, and stronger provenance, licensing, and content-safety controls.

20. Conclusion

The research supports a coherent architecture for an end-to-end AI video generation platform in which no single model is responsible for the entire production lifecycle. The strongest common design decision across the eight modules is separation of responsibilities connected through structured contracts and asynchronous orchestration.

Prompt Intelligence defines the user's intent. Planning turns that intent into executable dependencies. Image generation establishes visual references. Video generation supplies motion. QA evaluates whether generated media remains consistent with the intended identity, appearance, motion and semantics. Timeline rendering turns the approved assets into one synchronized artifact. Backend services persist state and enforce ownership, while infrastructure supplies the compute, storage, queueing and monitoring required to operate the platform.

The project should therefore be implemented as a measured, calibration-driven system. Exact model selection, thresholds, production hardware, provider pricing and commercial licensing remain implementation decisions. The research provides the architecture and candidate technologies; production validation must establish the actual performance and cost on the final workload.

Production-Level Model Selection & Cost Governance

Module	Primary production candidate	Fallback / alternative	Selection principle
1 Prompt Intelligence	Gemini 3.8 Flash + Jev	Clarification / alternative LLM	Reasoning + typed gates + cost/latency
2 Planning	Gemini 3.8 Flash + Jev	Selective re-plan / human review	Structured blueprint + DAG + fast decisions
3 Image	GPT - Image-2	Nano Banana 2 / SDXL	Reference conditioning + automation + cost
4 Video	Provider/model router	Veo / Wan / HunyuanVideo / LTX / CogVideoX	Project QA + VRAM + latency + license
5 QA & Accountability	AdaFace candidate + DINOv3 + RAFT + CLIP ; UUID+SHA-256 + Ed25519	MagFace / UniMatch / VLM	Diagnostic coverage + self-hosting
6 Render	GStreamer + FFmpeg + ffprobe	H.264/SVT-AV1 fallback	Deterministic media composition
7 Backend	FastAPI + PostgreSQL + Redis/Celery + S3	Managed equivalents	Stable contracts + async + media scaling
8 Infrastructure	Hybrid GPU/cloud	Fully managed or self-hosted	VRAM + concurrency + operational control

The eight modules are implemented as one production system. A model is selected only when it satisfies quality, latency, automation, reliability, licensing, data handling, reproducibility and cost-per-accepted-output requirements. The values below are planning references from the supplied module documents; final procurement must revalidate current pricing and licenses.

AI Video Platform - Infrastructure Components & Monthly Costs

#	Component	Full Form / Meaning	What It Does	Monthly Cost
1	EKS	Amazon Elastic Kubernetes Service	Manages containers; hosts FastAPI, Next.js, workers and services.	**$73**
2	ALB	Application Load Balancer	Receives web/API requests and distributes them to application servers/pods.	**$16**
3	WAF	Web Application Firewall	Inspects HTTP/HTTPS requests and blocks malicious traffic and abusive patterns.	**$15**
4	NAT Gateway	Network Address Translation Gateway	Allows private servers outbound internet access without public IPs.	**$66**
5	RDS PostgreSQL Multi-AZ	Amazon Relational Database Service + PostgreSQL + Multi-Availability Zone	Managed PostgreSQL with availability across Availability Zones.	**$350**
6	ElastiCache for Valkey	Amazon ElastiCache for Valkey	Fast in-memory data store/cache.	**$165**
7	S3	Amazon Simple Storage Service	Stores files and objects rather than database records.	**$23**
8	CloudFront	Amazon CloudFront — AWS Content Delivery Network (CDN)	Delivers files from locations closer to users.	**$112**
9	Secrets Manager	AWS Secrets Manager	Securely stores API keys, database passwords and credentials.	**$6**
10	KMS	AWS Key Management Service	Creates and manages encryption keys.	**$3**
11	ECR	Amazon Elastic Container Registry	Stores Docker/container images.	**$3**
12	CloudWatch	Amazon CloudWatch	AWS monitoring and logging service.	**$30**
13	AMP	Amazon Managed Service for Prometheus	AWS-managed Prometheus service for time-series metrics.	**$45**
14	Grafana	Grafana — visualization/observability platform	Turns metrics into dashboards and graphs.	**$18**
15	Temporal	Temporal Workflow Platform	Manages long-running stateful workflows, retries and failures.	**$100**

16	EKS CPU Application Nodes	Kubernetes CPU worker/application nodes	CPU servers running backend, frontend and non-GPU services.	**$400**
17	CPU Render Workers	CPU-based rendering/FFmpeg workers	Processes and assembles generated video clips using CPU resources.	**$200**
18	Backups	Database/storage backup infrastructure	Creates copies of important data for recovery.	**$25**

Baseline total (18 components): $1,650 / month

Production Acceptance Criteria Before Commercial Launch

Before commercial deployment, the following criteria shall be verified:

1. **System Reliability:** All modules shall have versioned input/output contracts, defined failure handling, monitoring, and controlled retry mechanisms.
2. **Model and Licensing Compliance:** All models, checkpoints, datasets, APIs, and third-party services shall be reviewed and approved for commercial use.
3. **Quality and Semantic Validation:** A benchmark dataset with human-validated results shall be used to evaluate image/video quality, prompt adherence, consistency, and semantic QA.
4. **Cost and Performance:** Cost per accepted video, including failed generations and retries, shall be measured. Load testing shall verify target concurrency, p50/p95 latency, queue time, and GPU utilization.
5. **Data, Consent and Provenance:** Data retention, security, residency, vendor policies, face consent, revocation.
6. **Traceability and Reproducibility:** Generated assets shall retain relevant prompt, model, settings, reference, and generation metadata to support auditing and reproducibility where technically supported.